Supplement A

The evolution of identity signals for coordination in diverse societies

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Appendix A Assortment Dynamics

Additionally, a homophily parameter, h, can be included to model contexts in which people interact more with those who broadcast the same social signals as themselves. Intuitively, the homophily parameter captures random assortment when $h = 0$ and assortment where agents who broadcast the same signal are twice as likely to interact as agents who broadcast differing signals when $h = 1$. Mathematically, this is captured by using the following utility function:

$$U(x) = \sum_{i \in Y} [H(h, x, i) \times M(i) \times p_{xi}] \qquad \text{if } x \neq i$$

$$U(x) = \sum_{i \in Y} [H(h, x, i) \times (M(i) - 1) \times p_{xi}] \qquad \text{if } x = i$$

where $H(h, x, i)$ is defined as:

$$H(h, x, i) = \frac{N}{\sum_{j \in Y} [S(h, x, j)]} \times S(h, x, i)$$

where N is the total number of agents in the population and $S(h, x, j)$ is:

$$S(h, x, j) = 2^h \qquad \text{if profile x entails broadcasting the same social signal as j}$$

$$S(h, x, j) = 1 \qquad \text{if x does not entail broadcasting the same social signal as j}$$

It is easy to see that this amounts to the same utility function as before when $h = 0$. Likewise, it is easy to see how $S(h, x, i)$ reflects agents with profile x having twice as much utility (when $h = 1$) from interactions with agents with profile i, due to more frequent interactions, when profiles x and i entail broadcasting the same social signal. The component of the equation requiring some explanation is $\frac{N}{\sum_{j \in Y}[S(h,x,j)]}$. This factor normalizes $S(h,x,i)$ relative to the signaling dispositions of the entire population.

To see why this normalization is necessary, consider counterfactually what would happen if we just defined $H(h,x,i)$ as equal to $S(h,x,i)$. Suppose further that for a population with the preferences of Table 5, every agent is broadcasting the same signal and playing their preferred action irrespective of the signal observed. If each type makes up half of the population, then agents are successfully coordinating in about half of their interactions. The signal being broadcast is meaningless for assortment since everyone is broadcasting the same signal. But suppose one agent mutates and starts broadcasting a different signal. Given the uniformity of everyone else in the population, the mutated agent is still just as likely to interact with a Bachite agent as a Stravinskian agent. Consequently, it will still be the cast that about half of the agent's interactions will result in successful coordination. However, with our counterfactual condition that $H(h,x,i)$ as equal to $S(h,x,i)$. The mutated agent's utility would suddenly be cut in half even though the agent is just as likely to successfully coordinate on her preferred action as before. But this makes no sense. By including the normalization factor $\frac{N}{\sum_{j \in Y}[S(h,x,j)]}$ in our definition of $H(h,x,i)$, we ensure that agents only see an increase in utility from broadcasting a particular signal in proportion to how much that signal impacts the frequency of their interactions.[1]

A.1 Simulating Assortment for One Dimensional Singals without the Attention Dynamics

Adding in assortment generally increases the probability of optimal outcomes in the model. This subsection shows simulation results for the same parameters as Section 3.2 in the main body of the paper.

Recall that, there are four different outcomes that frequently obtain in this model. In outcome (i) each type plays their preferred action all of the time is possible, but this outcome never occurs under the parameters shown in this section. Outcomes in which (ii) everyone plays Ummians' preferred action, and (iii) everyone plays Kishus' preferred action, are outcomes in which agents fail to make use of the social signals available to them. The outcomes in which (iv) Ummians (□) always play their preferred action and Kishus (⌂) play Umma greeting with Ummians (□) and play the Kish greeting among themselves, and

[1] If one wished to model a scenario in which an agent, in virtue of adopting an unused signal, simply ceased to interact with all agents since no other agents shared that signal, then it still would not make sense to let $H(h,x,i)$ equal $S(h,x,i)$. For that scenario, $S(h,x,i)$ should also be modified to be 0 when profile x does not entail broadcasting the same social signal as i. However, such antisocial behavior seems out of place given that this paper is modeling agents in coordination games where there is some common ground between agents of different types; i.e. an agents who prefers Bach still gets some benefit for coordinating on Stravinsky. In adopting the normalized equation, this paper reflects agents having the same number of interactions regardless of their signal, but preferentially interacting with agents who share their signal.

(v) Kishus (⌂) always play their preferred action and Ummians (□) play the Kish greeting with Kishus (⌂) and play the Umma greeting among themselves are optimal outcomes which depend on the agents evolving meaningful signals. Finally, though infrequent, (vi) there are a variety of suboptimal outcomes in which meaningful signals evolve but at least one type does not play their preferred action among themselves.

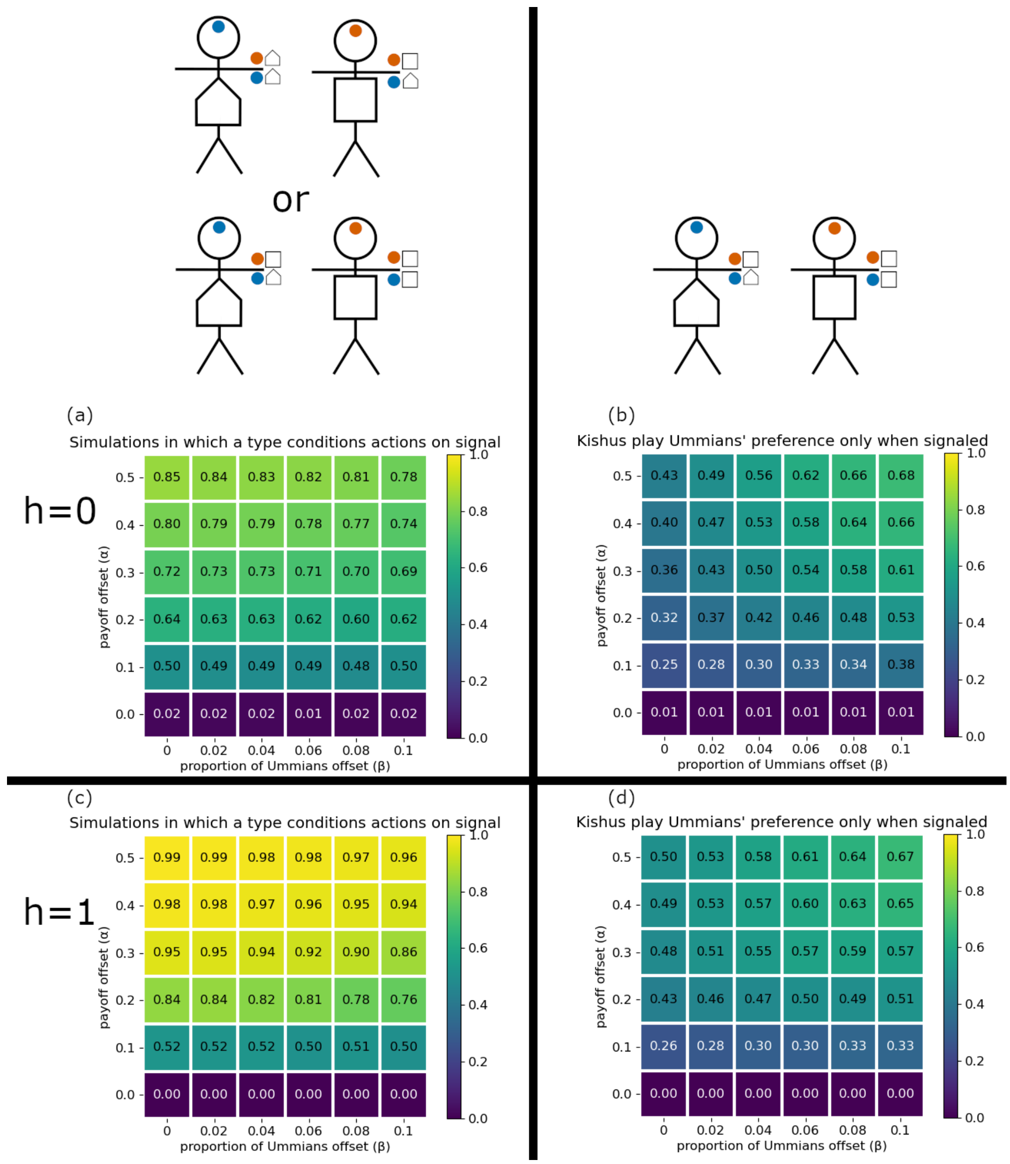


Figure A-1: (a) Proportion of outcomes that are (iv) or (v) for $h = 0$. (b) Proportion of outcomes that are (iv) for $h = 0$. (c) Proportion of outcomes that are (iv) or (v) for $h = 1$. (d) Proportion of outcomes that are (iv) for $h = 1$.

A.2 Simulating Assortment for One Dimensional Singals with the Attention Dynamics

At first glance, assortment appears to no longer promote optimal outcomes as the model becomes more complex. We are currently unsure of how to understand this. To some extent, this result seems to reflect it being the case that the assortment dynamics can incentivize signaling for a preference type that would otherwise avoid signaling (the Akkadians) during early stages of a simulation. When running a simulation with only as many signals as necessary for the optimal outcome to occur (2 signals), this results in agents getting stuck in a local optimum because an agent type that should adopt signals (the Ummians) has no signal available for them to adopt since both of the other agent types are already making use of the two available signals. While allowing agents more signals (Figure A-2 c & d) does mitigate the difference between outcomes from simulations with and with out assortment, it is still the case that optimal outcomes are more frequent without assortment (Figure A-2 c) than with assortment (Figure A-2 d).

Another explanation of could be related to transient diversity. In simulations where agents were afforded 3 signals, runs with no assortment (Figure A-2 c) took roughly 20% longer to reach stable equilibria than those with strong assortment (Figure A-2 d). We can also see that adding in assortment to the model dynamics does not immediately decrease the likelihood of the optimal outcome obtaining. Figure A-2 e shows simulations for weak assortment ($h = 0.5$) producing the optimal outcome within error of the results for the simulations with no assortment (A-2 c).

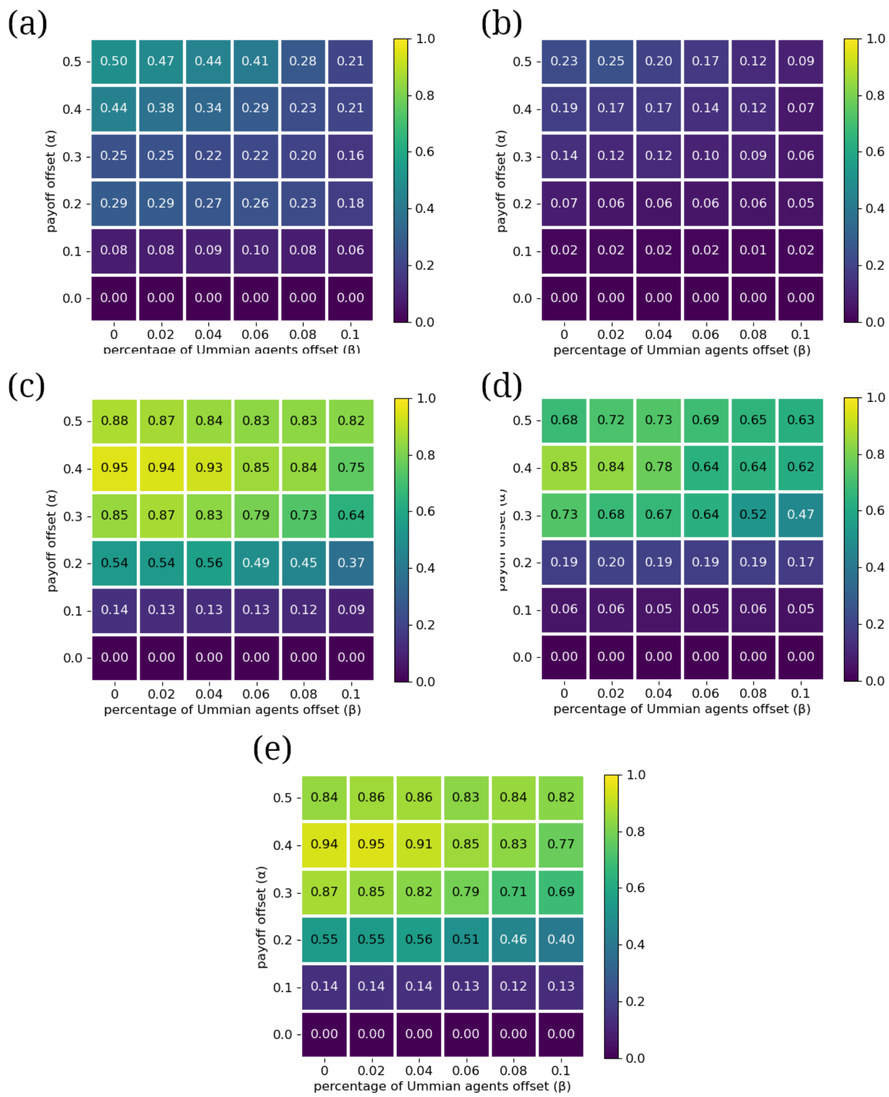


Figure A-2: Proportion of outcomes in which agents optimally condition their actions on signals when (a) afforded 2 signals with $h = 0$, (b) afforded 2 signals with $h = 1$, (c) afforded 3 signals with $h = 0$, (d) afforded 3 signals with $h = 1$, and (e) afforded 3 signals with $h = 0.5$.

A.3 Difficulties for Simulating Assortment for the Full Model with Multiple Dimensions of Signals

With our current code and the allowance of more signals than is necessary for the optimal outcome, simulating assortment for systems with multiple dimensions of social signals is not computationally tractable. We are working on a future model that uses different learning dynamics and we hope will resolve this issue of computational tractability. For now we have minimal simulation results to report (see Appendix B).

Appendix B Reinforcement Learning Dynamics

It is worth noting that nothing about the Generalized Bach or Stravinsky game requires agents to learn according to replicator dynamics. In this Appendix, we give some limited simulation results for the model in which the discrete replicator dynamics is replaced by agents learning according to Roth-Erev reinforcement learning with forgetting (Erev and Roth, 1998). A key benefit of reinforcement learning for future research is that it does not require quantifying the utility of an individual's strategy profile relative to the utility of all other profiles present in the subset of the population that shares that individual's coordination preferences. So, if one wanted to model a population whose preferences were evolving dynamically, the learning dynamics would function normally even if the dynamics lead to every individual in the population having unique coordination preferences.

Our implementation of reinforcement learning with forgetting can be thought of as each agent in the population having an urn for each location in the string representation of their strategy profile. Each urn begins a simulation with one ball corresponding to each possible action for that location in the strategy profile string. So, in the system that produces the single embedding structure, there is one non-null signal in each of two dimensions, leading to four total signal combinations, and three possible actions in response to each signal. So an agent would have six total urns. In the first urn a null ball and a blue ball, in the second a null ball and a red ball, and in the third through sixth urns a triangle, diamond, and star ball in each urn. In a single timestep, an agent begins by drawing one ball from each of their urns at random with equal probability. These draws then determine the strategy profile that the agent employs for the entire timestep while interacting with every other agent in the population. For each individual that is interacted with, if an agent receives a non-zero payoff the urns that contributed to their actions are reinforced. Reinforcements can be thought of as adding additional balls of the type that was drawn to the urn that it was drawn from. For example, suppose two Girshites both employing the strategy profile < ●●◇◇△✵ > successfully coordinate on the Girshite greeting (◇). Then, they would each add payoff-many blue balls to their first urn, payoff-many red balls to their second urn, and payoff-many diamond balls to their third urn. Since the urns corresponding to what action they choose in response to the signals ●_, _●, and __ were not used for this interaction, they are not reinforced. Finally, at the end of a timestep, forgetting is implemented by multiplying the number of balls of each type by a forgetting factor $0 < \phi < 1$. For any ball type whose quantity falls below 1 in any urn, that quantity is raised to 1; this helps preserve the possibility of that action being chosen in the future.

To represent assortment, we adjusted agents payoff by multiplying it by $H(h, x, i)$ where $H(h, x, i)$ is defined as:

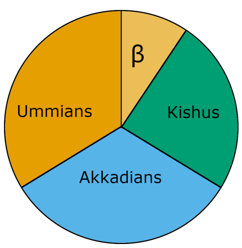
$$H(h, x, i) = \frac{N}{\sum_{j \in Y}[S(h, x, j)]} \times S(h, x, i)$$

where N is the total number of agents in the population and $S(h, x, j)$ is:

$$S(h, x, j) = (2 * d)^h \quad \text{if profile x entails broadcasting the same social signal as j in d-many dimensions}$$
$$S(h, x, j) = 1 \quad \text{if x does not entail broadcasting the same social signal as j}$$

We found that having agents select a single strategy profile that they employ for all interactions on a single timestep made this learning dynamic significantly more computationally tractable. We also suspect that it improves agents ability to learn in virtue of reducing the amount of noise in the system.

With these reinforcement learning dynamics, we ran simulations for a population of 500 agents where Lagashites were a $0.33 + \beta$ proportion of the population, Girshites were a $0.33 - \beta$ proportion of the population, Akkadians were a 0.34 proportion of the population, signal cost was -0.000125, and the forgetting factor, ϕ, was 0.998. The coordination preferences were:



Coordination Preferences	Lagash greeting	Girsu greeting	Akkadian greeting
Lagashites (△)	0.0075	0	0.00425
Girshites (◇)	0.00625	$0.0075 + \alpha$	0.00425
Akkadians (✵)	0	0	0.005

Table 1: Coordination preferences for the single embedding subsection.

This frequently produced the single embedding group structure. Note that, here again we see assortment making optimal outcomes more likely:

(a) (b)

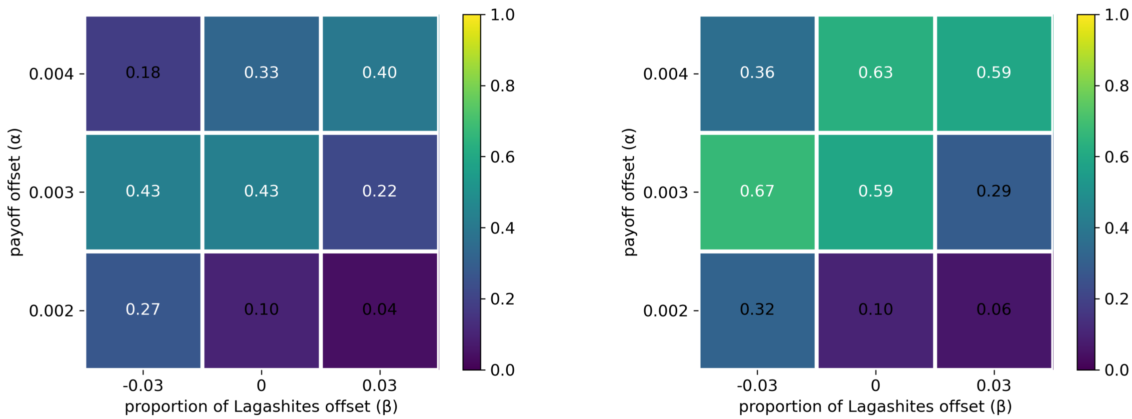


Figure A-3: the proportion of simulations that resulted in the optimal outcome, the single embedding group structure, when (a) $h = 0$ and (b) $h = 1$.

Appendix C Optimality Criteria

Coordination Preferences	Lagash greeting	Girsu greeting	Akkadian greeting
Lagashites (△)	1	0	0.5
Girshites (◇)	1	$1 + \alpha$	0.5
Akkadians (✲)	0	0	1

Table 2: Coordination preferences for the single embedding subsection of the main text.

Recall that in the main text, we designate a group structure as optimal if the following three conditions are met:

(i) Given any two agents, if there is some action for which they both receive nonzero payoff, then they always successfully coordinate on an action for which they both have nonzero payoff.

(ii) Given (i), that action provides the maximal possible sum of the two agents' payoffs.

(iii) There are no other group structures that meet criteria (i) & (ii) while also allowing agents to pay a lower attention cost.

It is straightforwardly apparent why conditions (i) and (iii) for our definition of an optimal outcome make the outcome desirable from the perspective of individual agents' incentives. Respectively, getting a payoff from successful coordination rather than not getting any payoff is desirable, and likewise paying minimal attention cost is desirable. But criterion (ii) can benefit from some explanation. The simplest explanation is that there are many strict Nash equilibria that we do not expect to see as outcomes from our simulations and which are obviously undesirable from the perspective of the agents' incentives. For example, lets call an outcome in which no agent in the population pays attention to signals and everyone gives the Akkadian greeting the 'All Akkadian' outcome. None of our simulations with the coordination preferences given in Table 2 resulted in the All Akkadian outcome. It is also a clearly undesirable outcome as Lagashites would have higher payoffs if they were coordinating on the Lagashite greeting among themselves, and likewise for the Girshites among themselves. Nonetheless, the All Akkadian outcome satisfies criteria (i) and (iii) and is a strict Nash equilibrium. Criterion (ii) eliminates the All Akkadian outcome from being considered optimal. It is not the only possible criterion that could accomplish this. For example, consider the following alternative criterion:

(ii) *alternative* Given (i), that action is such that there is no other action for which both agents receive a strictly greater payoff.

This alternative criterion (ii) selects the same outcomes as optimal for all of the coordination preferences that are considered in this paper's main text, so we do not have any reason to prefer one criterion over the other. However, we can illustrate coordination preferences for which the two criteria do differ. Consider coordination preferences for a slightly more complex system where there are two Lagash greetings, A and B:

(a) Coordination preferences in which we find the first criterion (ii) more appropriate:

Coordination Preferences	Lagash greeting A	Lagash greeting B	Girsu greeting	Akkad greeting
Lagashites	1.5	1	0	0.5
Girshites	1	1	1.5	0.5
Akkadians	0	0	0	1

(b) Coordination preferences in which we think the alternative criterion (ii) may be more appropriate:

Coordination Preferences	Lagash greeting A	Lagash greeting B	Girsu greeting	Akkad greeting
Lagashites	1.5	1	0	0.5
Girshites	1	1.25	1.5	0.5
Akkadians	0	0	0	1

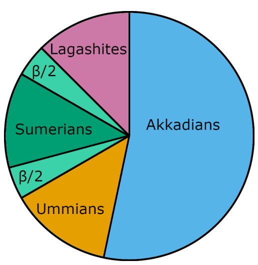
For both of these sets of coordination preferences, our first criterion (ii) will *only* select an outcome as optimal if Girshites and Lagashites coordinate with each other on the Lagash greeting A. But our alternative criterion (ii), for both of these sets of coordination preferences, will consider as optimal outcomes in which Lagashites and Girshites coordinate on *either* the Lagash greeting A *or* the Lagash greeting B. But, our intuition based on what we have seen of how agents strategy profiles evolve in the model, is that for the (a) coordination preferences outcomes in which Lagashites and Girshites coordinate on the Lagash greeting B would be rare and for the (b) coordination preferences such outcomes would be common. In the (a) coordination preferences, Girshites are indiferent between the two Lagash greetings while Lagashites strongly prefer the A greeting; so even in a model that evolves only according to the incentives of individual agents, it seems like it would take a very bespoke evolutionary path to result in Lagahsites and Girshites coordinating on the Lagash greeting B. However, this reasoning goes out the window for the (b) coordination preferences because now the Girshites are no longer indifferent between the two Lagash greetings, instead having a substantive preference for the B greeting. It is possible that our intuitions are mistaken here since we have not done a thorough investigation of the (a) and (b) coordination preferences. But from a perspective of individual agents' incentives we think it is at the very least a non trivial question to ask what is the best criterion for determining an optimal outcome in such situations. Luckily, for the coordination preferences considered in the main text, it makes no difference.

We think it is possible that rather than choosing between either of the criteria at hand, some equilibrium notion may be appropriate for picking out optimal outcomes in future research. There are two equilibrium notions that may be relevant, though we are uncertain of how fruitful pursuing them may be. The notion of a strong Nash equilibrium seems possible useful if we consider the All Akkadian outcome and how it can become unstable when agents are exploring a sufficient amount. A single agent unilaterally deviating from her strategy profile in the All Akkadian outcome will have a strictly lower payoff than if she had not deviated. However, two agents who are either Lagashite or Girshite could together improve their payoff by adopting the same social signal and upon seeing that social signal coordinating on an action for which they both had a higher payoff on than the Akkadian greeting. Thus while the All Akkadian outcome is a strict Nash equilibrium, it is not a strong Nash equilibrium. On the other hand, using the concept of a strong Nash equilbrium seems infelicitous because of its assumption of unlimited private communication between agents. This is especially problematic for a model in which we are exploring how social signaling (a type of public communication) evolves. The notion of a trembling hand perfect equilibrium also seems promising, but we are uncertain of its efficacy for an extensive form game played by a large population (and the extensive form is certainly the appropriate formalization of the game once social signals are involved). We additionally worry that the mere existence of a sequence of perturbed games leading to a particular outcome may be too permissive; i.e. we think it likely that in some explication of a trembling hand perfect equilibrium for a large population playing an extensive form game would call optimal an outcome in which Lagashites and Girshites coordinate on the Lagashite B greeting while having the (a) coordination preferences. Thus trembling hand perfect may require further refinement to fit our context. Further analysis of these equilibrium notions is beyond the scope of the present paper.

Appendix D The Disjoint Double Embedding

This appendix has a data from a broader range of simulation parameters for producing the disjoint double embedding structure. It's main purpose is to show the rationale behind using stronger mutation values in the learning dynamics for the results that were reported in the main body of the paper. Up until the section on double embeddings, mutation rates were $m_0 = 0.01$ and $m_1 = 0.1$. However the double embedding section shows simulation results for $m_0 = 0.05$ and $m_1 = 0.2$. The short answer explaining this change is just that in the more complex parameter space of the system that produces the disjoint double embedding, having agents explore different possibilities more frequently is very beneficial. At the end of this appendix, we also point out the possibility of a distinct type of disjoint double embedding that could occur in a population. Though the difference between these two types of disjoint double embeddings is narrow.

Recall that for this system, coordination preferences are:



Coordination Preferences	Sumerian greeting	Umma greeting	Lagash greeting	Akkadian greeting
Sumerians (○)	1	0	0	0.25
Ummians (□)	1	$1+\alpha$	0	0.25
Lagashites (△)	1	0	$1+\alpha$	0.25
Akkadians (✲)	0	0	0	1

Sumerians (○) make up $0.15+\beta$ proportion of the population, Ummians (□) are a $0.15-0.5\times\beta$ proportion of the population, Lagashites (△) are a $0.15-0.5\times\beta$ proportion of the population, and Akkadians (✲) are a 0.55 proportion of the population. There is one signal in the first dimension and two in the second.

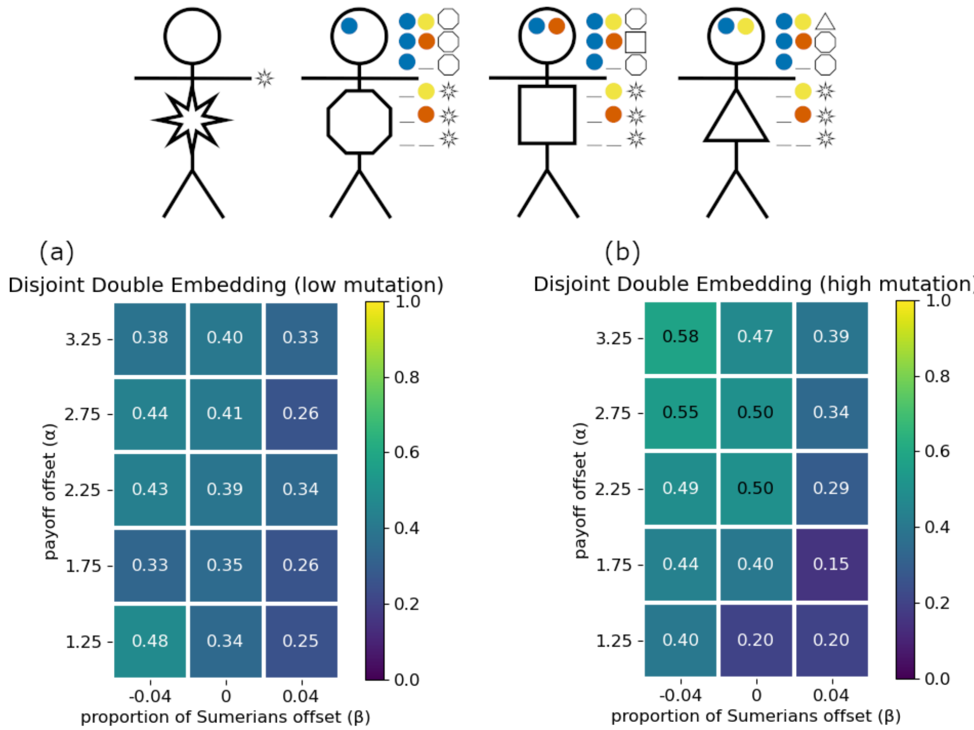


Figure A-4: Proportion of outcomes that are the disjoint double embedding structure when (a) the mutation rate is $m_0 = 0.01$ and $m_1 = 0.1$, or (b) the mutation rate is $m_0 = 0.05$ and $m_1 = 0.2$.

Figure A-4 shows results using the same mutation rate as prior sections, $m_0 = 0.01$ and $m_1 = 0.1$, as well as results with a higher mutation rate, $m_0 = 0.05$ and $m_1 = 0.2$. In general, it seems learning is more effective with higher mutation rates as the learning context become more complex. This also requires using higher payoffs to cut through the noise of the increased mutation rate. However, it is difficult to make these claims with certainty as these more complex systems are less computationally tractable, which precludes the more robust parameter sweeps that were done with simpler models. Still there is a rationale for why higher mutation rates might be helpful for learning in the more complex systems. The system that produces the single embedding structure in Section 4.2 only has 324 different possible strategy profiles. Thus, we can expect a large portion of the different strategy profiles to be present on the first timestep of a simulation when they are assigned. Mutation is most likely not needed to produce an agent with any given strategy profile. However, the system in this section allows 24,576 different possible strategy profiles. This means that for a population of 1,000 agents, on any given timestep, there is at most around 4% of the different strategy profiles present in the population. Consequently, a higher mutation rate is likely beneficial in virtue of allowing agents to explore a broader swath of the possible strategy profiles.

Just as with the single embedding structure, this section on the disjoint double embedding structure does not illustrate any key point about the model. Rather it is just a further illustration of how the model works. At the beginning of a simulation, as always, agents are randomly assigned social signals to broadcast or not and are randomly assigned greetings to give in response to the social signals. What is not random is the preferences that we assign to the agents. Since Ummians (□) and Lagashites (△) have a preference for coordinating on the Sumerian greeting (○) over the Akkadian greeting they frequently both adopt the social signal that comes to be associated with the Sumerian identity. Their even stronger preference for their own cities' greetings and the fact that they both get no payoff from the other's city greeting produces the incentive for the Ummians and Lagashites to adopt social signals in a second dimension that are distinct from each other and allow them to successfully coordinate on their most preferred greeting among others from their own city. All of this comes together as what we call the disjoint double embedding. The fact that the Ummians and Lagashites had a deep seated rivalry is important to how we've set up this disjoint double embedding structure. Not only is it the justification for Ummians and Lagashites getting no payoff from each other's greetings, but it explains why it is appropriate to allow two signals in the same dimension (which are therefore mutually exclusive) rather than having an additional dimension for each of the signals that the Ummians and Lagashites adopt in the disjoint double embedding structure (see Figure A-5).

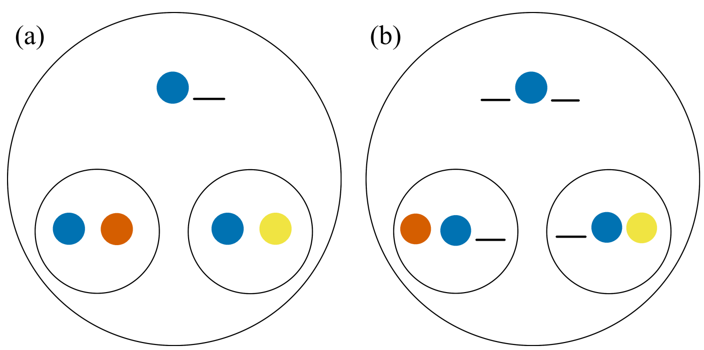


Figure A-5: The left side (a) depicts the signaling system that we see showed results for in the disjoint double embedding structure, where there are just two dimensions of social signals with the second dimension having two mutually exclusive social signals. The right side (b) shows an alternative possible double embedding in which the red and yellow signals are in different dimensions and therefore not mutually exclusive.

Appendix E Analysis of Section 2.3 Model

Recall that Section 2.3 gave the mean outcome of 1,000 simulations of the generalized BoS game with a population of 1,000 agents, $m_0 = .01$, $m_1 = .1$, and run for more than a sufficiently large number of time steps to reach an equilibrium (4×10^4).[2] Type 0 agents make up $0.5 + \beta$ proportion of the population (and type 1 agents are a $0.5 - \beta$ proportion). Coordination preferences are:

Coordination Preferences	Bach	Stravinsky
type 0	$1 + \alpha$	1
type 1	1	$1 + \alpha$

Table 3: Coordination Preferences: generalized BoS for a population of two types, $\alpha \geq 0$.

Recall that agents begin the simulation with randomly selected strategy profiles. Thus we can assume that roughly half of the agents play Bach and half play Stravinsky on the first timestep of a simulation. This should result in almost all of the agents playing their preferred action on the second time step. We can see this by first calculating the utilities:

$$\begin{aligned} U(\text{preferred action}) &= 5 \times 10^2 \times (1 + \alpha) \\ U(\text{undesired action}) &= 5 \times 10^2 \times 1 \end{aligned}$$

Then we can calculate the change in the population $N(\text{preferred action})$ as:

$$\begin{aligned}
= &\quad 2.5 \times 10^2 + 2.5 \times 10^2 \times [5 \times 10^2 \times (1+\alpha) - 0.5 \times (5 \times 10^2 \times (1+\alpha) + 5 \times 10^2)] \\
= &\quad 2.5 \times 10^2 + 2.5 \times 10^2 \times [5 \times 10^2 \times (1+\alpha) - 0.5 \times (5 \times 10^2 \times (1+\alpha) + 5 \times 10^2 \times \frac{1+\alpha}{1+\alpha})] \\
= &\quad 2.5 \times 10^2 + 2.5 \times 10^2 \times 2.5 \times 10^2 \times (1+\alpha) \times [2 - (1 + \frac{1}{1+\alpha})] \\
= &\quad 2.5 \times 10^2 + 2.5 \times 10^2 \times 2.5 \times 10^2 \times (1+\alpha) \times (1 - \frac{1}{1+\alpha} \\
= &\quad 2.5 \times 10^2 \times (1 + 2.5 \times 10^2 (1+\alpha) - 2.5 \times 10^2) \\
= &\quad 2.5 \times 10^2 \times (1 + 2.5 \times 10^2 \times \alpha)
\end{aligned}$$

and $N(\text{preferred action})$ is the same with $\alpha = 0$ therefore the proportion of

[2] In these simulations, every 100 timesteps it was checked whether the distribution of agents' strategy profiles was unchanged. If so, the simualtion was halted prior to reaching 4×10^4 timesteps.

agents of a given type that play their preferred action should be:

$$
\begin{aligned}
& \frac{N(\text{preferred action})}{N(\text{preferred action}) + N(\text{undesired action})} \\
= & \frac{2.5 \times 10^2 \times (1 + 2.5 \times 10^2 \times \alpha)}{2.5 \times 10^2 \times (1 + 2.5 \times 10^2 \times \alpha) + 2.5 \times 10^2} \\
= & \frac{1 + 2.5 \times 10^2 \times \alpha}{1 + 2.5 \times 10^2 \times \alpha + 1}
\end{aligned}
$$

which is close to 1 for the relevant values of α. In other words, we should expect almost all of the agents to be playing their preferred action after the first time step. In accordance with this, when the the mutation rates were set to zero, all simulations resulted in the outcome where all agents play their types preferred action.

Now, if we assume all of the agents are playing their preferred action, then we can calculate when type 1 agents', whose preferred action is Stravinsky, should have a higher utility for playing Stravinsky than Bach:

$$
\begin{aligned}
U(< S >) &> U(< B >) \\
(0.5 - \beta) \times 10^3 \times (1 + \alpha) &> (0.5 + \beta) \times 10^3 \\
0.5 - \beta + \alpha \times (0.5 - \beta) &> (0.5 + \beta) \\
\alpha \times (0.5 - \beta) &> (0.5 + \beta) - (0.5 - \beta) \\
\alpha &> \frac{2\beta}{0.5 - \beta}
\end{aligned}
$$

Figure A-6 shows the graph of $\alpha = \frac{2\beta}{0.5-\beta}$ and it is easy to see that this aligns with the simulation results shown in Figure 3.

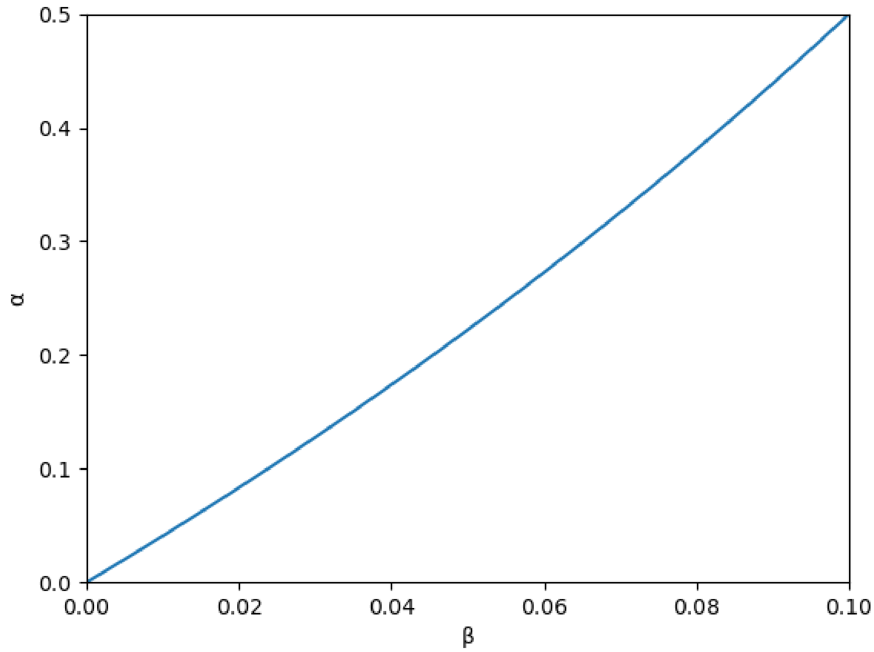


Figure A-6: Type 1 agents should continue playing their preferred action if α and β place them above this line.

Appendix F Increasing Population Size Decreases Variance in Outcomes

This appendix compares data points from Sections 2.3 & 3.4 with data points that use the same parameters but increase to population size from 1,000 agents to 10,000 agents. Here we see that there is less variance in outcomes for larger populations; the explanation for this is straightforward and given below.

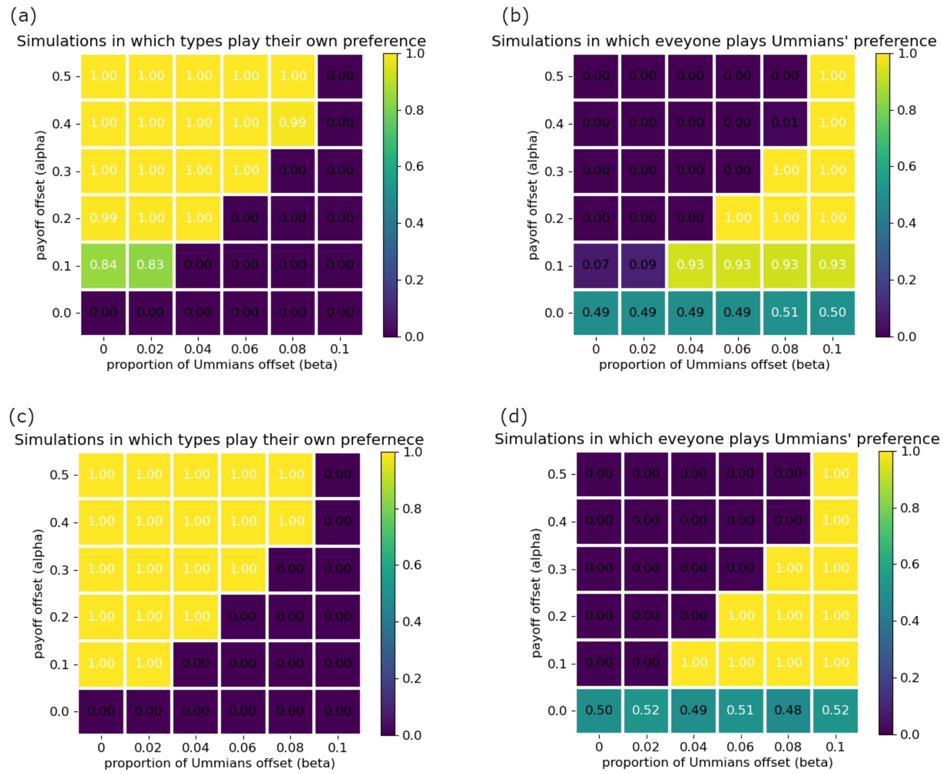


Figure A-7: (a)Population size = 1,000 agents. Proportion of simulations that resulted in outcome (i) in which agents always give their preferred greeting; this means that when agents of different types fail to coordinate. (b) Population size = 1,000 agents. Proportion of simulations that resulted in outcome (ii)in which everyone gives Ummians' preferred greeting.(c)Population size = 10,000 agents. Proportion of simulations that resulted in outcome (i) in which agents always give their preferred greeting; this means that when agents of different types fail to coordinate. (d) Population size = 10,000 agents. Proportion of simulations that resulted in outcome (ii)in which everyone gives Ummians' preferred greeting.

Comparing Figures A-7a&b with A-7c&d we see that there is more variance in the outcomes for simulations of smaller populations than larger populations. This can be understood by considering what must happen to obtain outcome (iii), everyone plays the Kish greeting. This is the same as asking when is it the case that agents' playing the Kish greeting have higher utility from that action than agents playing the Umma greeting. Note that since Kishu agents prefer the Kish greeting over the Umma greeting and vice versa for Ummian agents, then an Ummian agent having higher utility for playing the Kish greeting than the Umma greeting implies that Kishu agents will also have higher utility for playing the Kish greeting than the Umma greeting. So, lets just consider when an Ummian agent has higher utility for playing the Kish greeting than the Umma greeting. By the $U(x)$ equation this happens when:

$$\begin{aligned} U(K) &> U(U) \\ M(K) \times 1 &> M(U) \times (1+\alpha) \end{aligned}$$

where $M(x)$ is the number of agents playing x. Suppose for illustration that $\alpha = 0.1$. Then if the population size is 1,000 (i.e. $M(K)+M(U) = 1000$) agents we get that outcome (iii) is likely to occur if:

$$\begin{aligned} M(K) \times 1 &> M(U) \times 1.1 \\ 1000 - M(U) &> M(U) \times 1.1 \\ \frac{1000}{1.1M(U)} - \frac{M(U)}{1.1M(U)} &> 1 \\ \frac{1000}{1.1M(U)} - \frac{10}{11} &> 1 \\ \frac{10000}{11M(U)} &> \frac{21}{11} \\ 10000 &> 21M(U) \\ 476.19047619 &> M(U) \end{aligned}$$

Similarly, one could calculate that for a population of 10,000 agents the number of agents playing the Umma greeting must be less than 4761.9047619 for Ummian agents to have higher utility from playing the Kish greeting than the Umma greeting. Now if we consider just the first timestep of a simulation in which agents are randomly (with equal probability) assigned strategy profiles, then it is easy to check that it is more likely that (a), a population of 1000 agents will have 476 or fewer agents playing the Umma greeting (and 524 or more playing the Kish greeting), than it is for (b) a population of 10,000 agents to have 4761 or fewer agents playing the Umma greeting (and 5239 or more playing the Kish greeting); i.e. the probability of (a) is:

$$\sum_{k=0}^{476} \binom{1000}{k} \times 0.5^{1000} \approx 0.06858400263532671$$

and the probability of (b) is:

$$\sum_{k=0}^{4761} \binom{10000}{k} \times 0.5^{10000} \approx 0.00000091718051570$$

While these calculations only apply to the probability of Ummian agents having higher utility for playing the Kish greeting than the Umma greeting *on the first* *timestep*. They should provide an intuition for the general case of why low probability outcomes are more likely for smaller populations than larger ones.

We can also see this effect of increasing population size for the model in Section 3.4:

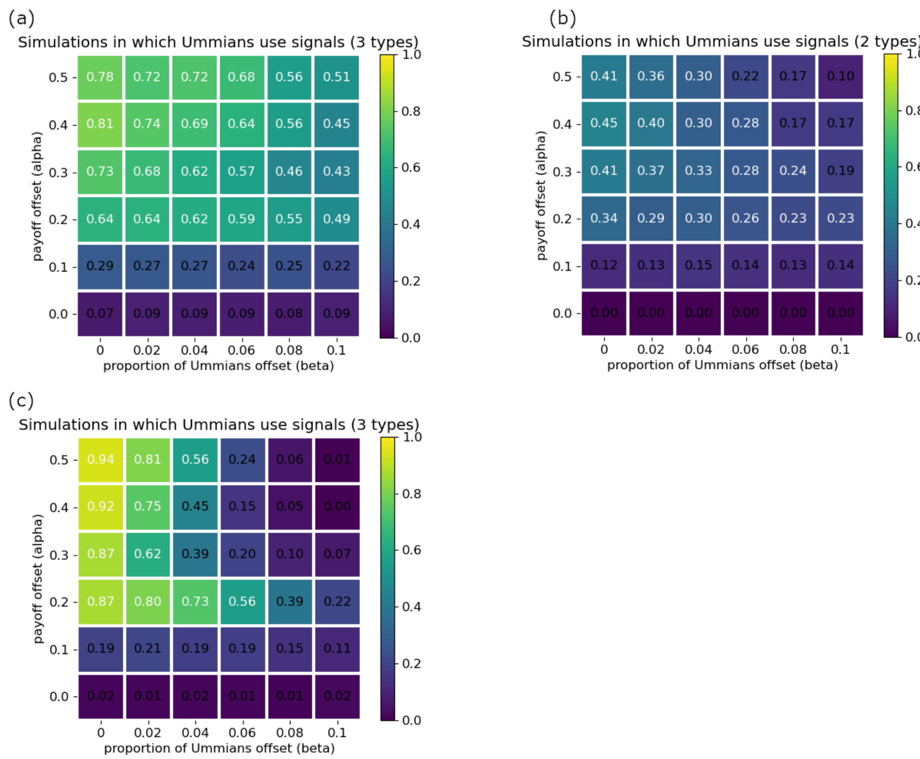


Figure A-8: (a) Proportion of outcomes in which type 0 agents broadcast a social signal, for the model with three types of agents, 1,000 total agents. (b) Proportion of outcomes in which type 0 agents broadcast a social signal, for the model with two types of agents, 1,000 total agents. (c) Proportion of outcomes in which type 0 agents broadcast a social signal, for the model with three types of agents, 10,000 total agents.

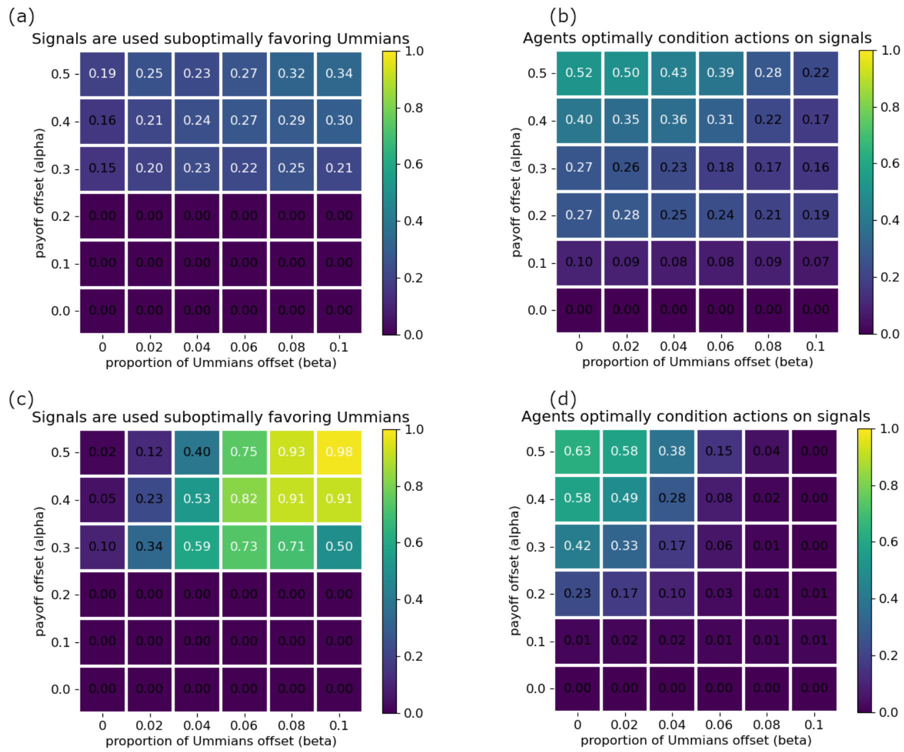


Figure A-9: (a) Proportion of outcomes that are (viii), population of 1,000 agents. (b) Proportion of outcomes that are (xii-xiii), population of 1,000 agents. (c) Proportion of outcomes that are (viii), population of 10,000 agents. (d) Proportion of outcomes that are (xii-xiii), population of 10,000 agents.

Appendix G Additional Details on Simulation Results for Generalized BoS with Attention and Signal Costs

Note: this appendix uses data that was generated independently of the data points given in Appendix F. Therefore, there may be nominal differences in the values given in corresponding heatmaps though the general trends remains the same. This appendix's simulations were done with a population size of 10,000 agents; so it can be check that Figures A-12 and A-14 closely match Figures A-8c and A-9d from Appendix F.

Additionally, note that a supplemental PDF shows the evolution of signaling and behavioral dispositions of individual simulation runs. For each of the seventeen outcomes here, ten different simulations resulting in the given outcome are detailed except when fewer than ten simulations resulted in the outcome.

The supplemental PDF was composed prior to developing the Mesopotamian greetings example. It was written in terms of agents deciding between Bach, Stravinsky, or EDM rather than being in terms of the Umma greeting, the Kish greeting, and the Akkadian greeting. As I am disinclined to redo the 116 pages of the supplemental PDF, this appendix is written in terms of Bach, Stravinsky, and EDM. My appologies for the change. Bellow, the table of coordination preferences is restated in those terms with the corresponding Mesopotamian greetings given in parentheses.

Finally, it should be noted that the outcomes listed here are relative to the most frequent behavior of the agents rather than the behavior of all of the agents. For outcomes (i)-(xiii) these two measures of an outcome are equivalent. However for outcomes (xiv)-(xvii), the outcomes are what **?** *call hybrid equilibria, though they now prefer the bipartite signaling equilibria as the term "hybrid equilibria" has previously been used to refer to something else. An example is illustrated here for outcome (xiv), Figure A-16, and details of the other outcomes can be found in the supplemental PDF. All of these bipartite signaling equilibria were confirmed to be Nash equilibria.*

Coordination Preferences	Bach (Umma)	Stravinsky (Kish)	EDM (Akkadian)
type 0	$1+\alpha$	1	0.5
type 1	1	$1+\alpha$	0.5
type 2	0	0	1

For the parameters investigated, there were thirteen different outcomes that occurred at least 0.5% of the time for some data point.

(i) Outcomes in which there was no coordination between agents of different types, i.e. agents always play their preference irrespective of the signal transmitted. This is shown in Figure A-10.

(ii–ix) Agents exhibit suboptimal coordination, but in an intuitive way. This is shown in Figure A-11.

> (viii) A notably common outcome was one in which type 0 and type 2 agents always played their respective preference, and type 1 agents played B with type 0, S with type 1, and EDM with type 2. The prevalence of this outcome is shown in Figure A-12. This outcome was frequently characterized by type 0 agents signaling 0 and type 1 and 2 agents attending to signals that reliably identified their type. One can check that this is a Nash equilibrium. It is suboptimal in the sense that when type 0 agents are paired with type 2 agents, there is a failure to coordinate.

(x–xi) Agents condition their actions suboptimally, and in a counterintuitive way. This is shown in Figure A-13. Specifically what was counterintuitive about these outcomes is that they involved a type playing their preference when paired with a type other than themselves, but not playing their preference among themselves.

(xii–xiii) Agents condition their actions optimally, in the sense that there were never failures of coordination. This is shown in Figure A-14.

(xii) A notably common outcome was one in which type 0 played their preference when paired with type 1. The prevalence of this outcome is shown in Figure A-15.

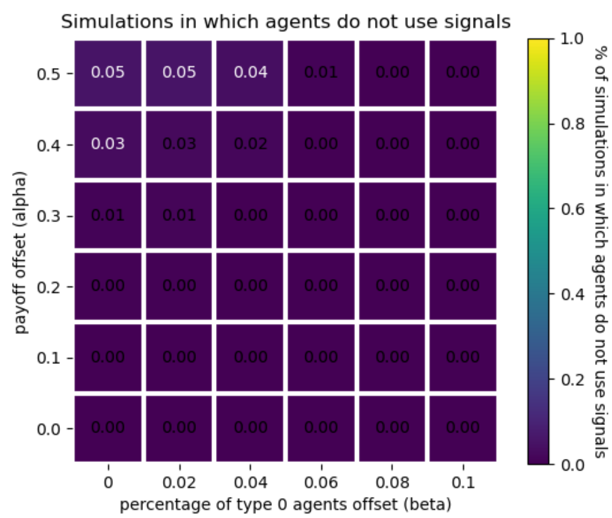


Figure A-10: Proportion of outcomes that are (i).

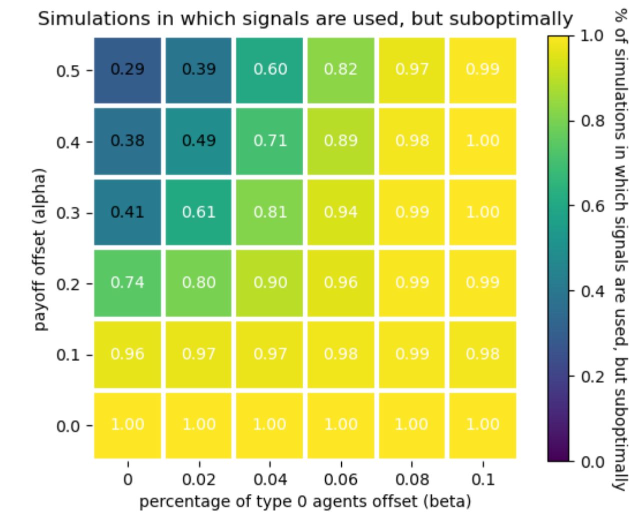


Figure A-11: Proportion of outcomes that are (ii-ix).

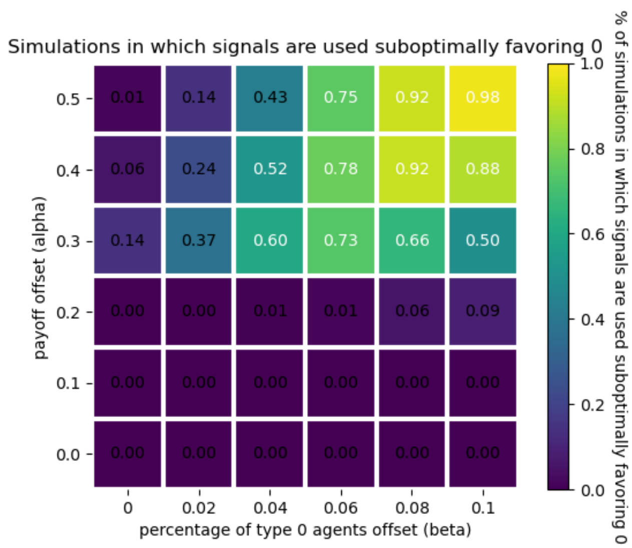


Figure A-12: Proportion of outcomes that are (viii).

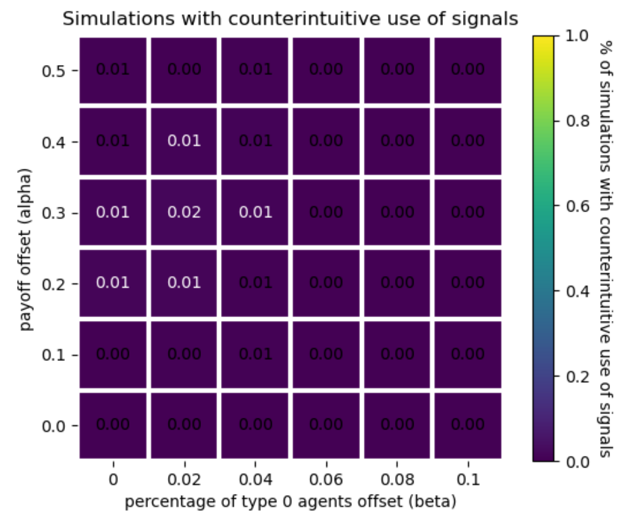


Figure A-13: Proportion of outcomes that are (x-xi).

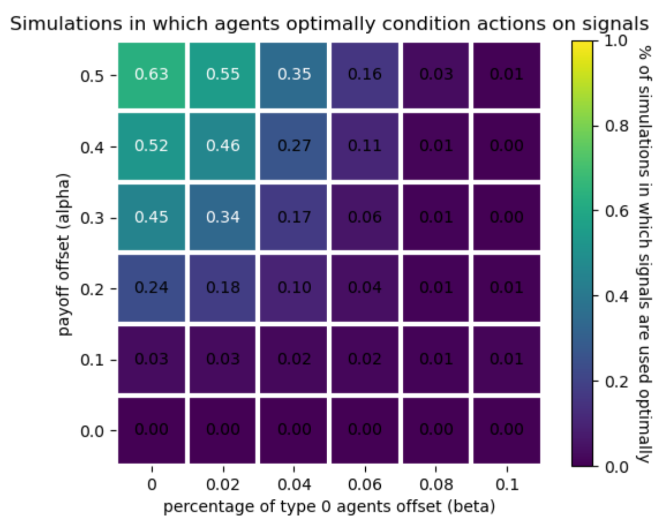


Figure A-14: Proportion of outcomes that are (xii-xiii).

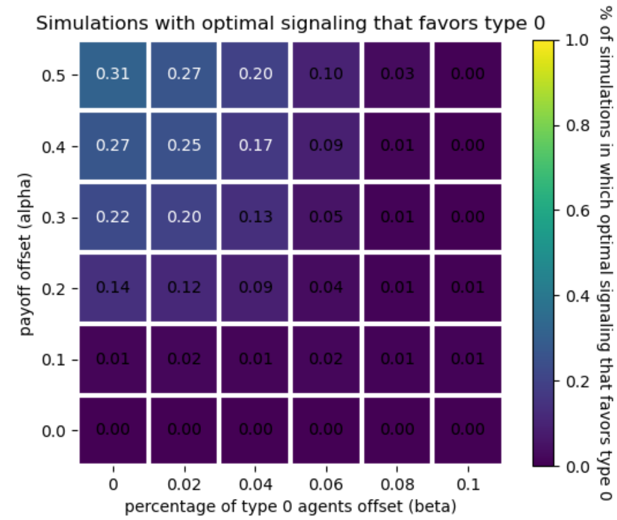


Figure A-15: Proportion of outcomes that are (xii).

outcome (i)

y plays this with x	type 0	type 1	type 2
type 0	B	B	B
type 1	S	S	S
type 2	E	E	E

outcome (ii)

y plays this with x	type 0	type 1	type 2
type 0	B	B	B
type 1	B	B	B
type 2	E	E	E

outcome (iii)

y plays this with x	type 0	type 1	type 2
type 0	S	S	S
type 1	S	S	S
type 2	E	E	E

outcome (iv)

y plays this with x	type 0	type 1	type 2
type 0	B	B	E
type 1	B	B	E
type 2	E	E	E

outcome (v)

y plays this with x	type 0	type 1	type 2
type 0	S	S	E
type 1	S	S	E
type 2	E	E	E

outcome (vi)

y plays this with x	type 0	type 1	type 2
type 0	B	B	B
type 1	B	S	B
type 2	E	E	E

outcome (vii)

y plays this with x	type 0	type 1	type 2
type 0	B	S	S
type 1	S	S	S
type 2	E	E	E

outcome (viii)

y plays this with x	type 0	type 1	type 2
type 0	B	B	B
type 1	B	S	E
type 2	E	E	E

outcome (ix)

y plays this with x	type 0	type 1	type 2
type 0	B	S	E
type 1	S	S	S
type 2	E	E	E

outcome (x)

y plays this with x	type 0	type 1	type 2
type 0	S	B	E
type 1	B	S	E
type 2	E	E	E

outcome (xi)

y plays this with x	type 0	type 1	type 2
type 0	B	S	E
type 1	S	B	E
type 2	E	E	E

outcome (xii)

y plays this with x	type 0	type 1	type 2
type 0	B	B	E
type 1	B	S	E
type 2	E	E	E

outcome (xiii)

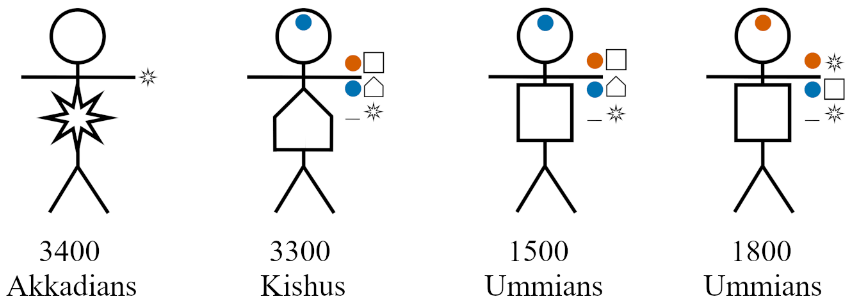
y plays this with x	type 0	type 1	type 2
type 0	B	S	E
type 1	S	S	E
type 2	E	E	E

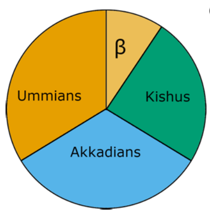
outcome (xiv)

y plays this with x	type 0	type 1	type 2
type 0	B	B	B
type 1	B	E	E
type 2	E	E	E

This occurred:
1 time(s) when alpha = 0.1 and beta = 0.08
2 time(s) when alpha = 0.2 and beta = 0.08
1 time(s) when alpha = 0.2 and beta = 0.1

Here's an example of what one of these outcomes looked like:





outcome (xv) α=0.3 β=0

Coordination Preferences	Umma greeting	Kish greeting	Akkadian greeting
Ummians (□)	$1+\alpha$	1	0.5
Kishus (⌂)	1	$1+\alpha$	0.5
Akkadians (✲)	0	0	1

Figure A-16: In this bipartite signaling equilibrium, the population of the Kishus is split between two different ways of signaling and conditioning actions on signals. It can be checked that this is a Nash equilibrium, i.e. no agent can unilaterally change their behavior in a way that increases their net payoff.

outcome (xv)

y plays this with x	type 0	type 1	type 2
type 0	E	B	E
type 1	B	S	E
type 2	E	E	E

This occurred:
1 time(s) when alpha = 0.3 and beta = 0
1 time(s) when alpha = 0.4 and beta = 0

outcome (xvi)

y plays this with x	type 0	type 1	type 2
type 0	B	S	E
type 1	S	E	E
type 2	E	E	E

This occurred:
1 time(s) when alpha = 0.3 and beta = 0.04
2 time(s) when alpha = 0.4 and beta = 0.02
1 time(s) when alpha = 0.5 and beta = 0
1 time(s) when alpha = 0.5 and beta = 0.02
1 time(s) when alpha = 0.5 and beta = 0.04

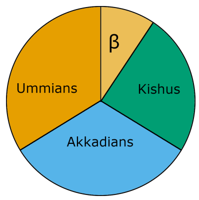
outcome (xvii)

y plays this with x	type 0	type 1	type 2
type 0	B	E	E
type 1	E	S	E
type 2	E	E	E

This occurred:
1 time(s) when alpha = 0.3 and beta = 0.06
1 time(s) when alpha = 0.4 and beta = 0.08
1 time(s) when alpha = 0.5 and beta = 0.06
2 time(s) when alpha = 0.5 and beta = 0.1

Appendix H Examining Initial Conditions in the One Dimensional Model with Attention and Signal Costs

In the main text, all simulations are performed with a population whose initial strategy profiles are randomly assigned to agents prior to the first timestep. Here, we consider the model from Section 3.4, but with the amendment that initial strategy profiles are not randomly assigned to agents. We consider two possible non-random assignments of initial strategy profiles (i) no agents signals and every agent plays their most preferred greeting, and (ii) no agent signals and every agent plays the Akkadian greeting. Given this change, it was rare (less than 1% of the time) to see optimal outcomes using the same parameters as the main text. Here we show simulation results in which the payoff offset (α) ranges from 0 to 1.5 (as opposed to ranging from 0 to 0.5 in the main text) and in which the mutation rates are $m_0 = 0.05$ and $m_1 = 0.2$ (as opposed to $m_0 = 0.01$ and $m_1 = 0.1$ in the main text). All other parameters remain identical with the main text. Additionally, we show simulation results from the main text's model, in which agents are initialized with a random strategy profile, using these new α, m_0, and m_1 values. As a reminder, these are the coordination preferences for these simulations:



Coordination Preferences	Umma greeting	Kish greeting	Akkadian greeting
Ummians (□)	$1+\alpha$	1	0.5
Kishus (⌂)	1	$1+\alpha$	0.5
Akkadians (✵)	0	0	1

Table 4: Coordination payoffs for the model with three preference types.

Simulation results for the proportion of simulations that resulted in one of the two optimal outcomes are shown in Figure A-17. The simplest explanation of why the higher payoff values and mutation rates are needed to see optimal outcomes is that this is a consequence of the initial conditions (both (i) and (ii)) being a Nash equilibriums that are relatively stable though suboptimal. Thus agents benefit immensely from increased exploration (higher m_0, and m_1 values) and stronger incentive to pursue their preferences (higher α values).

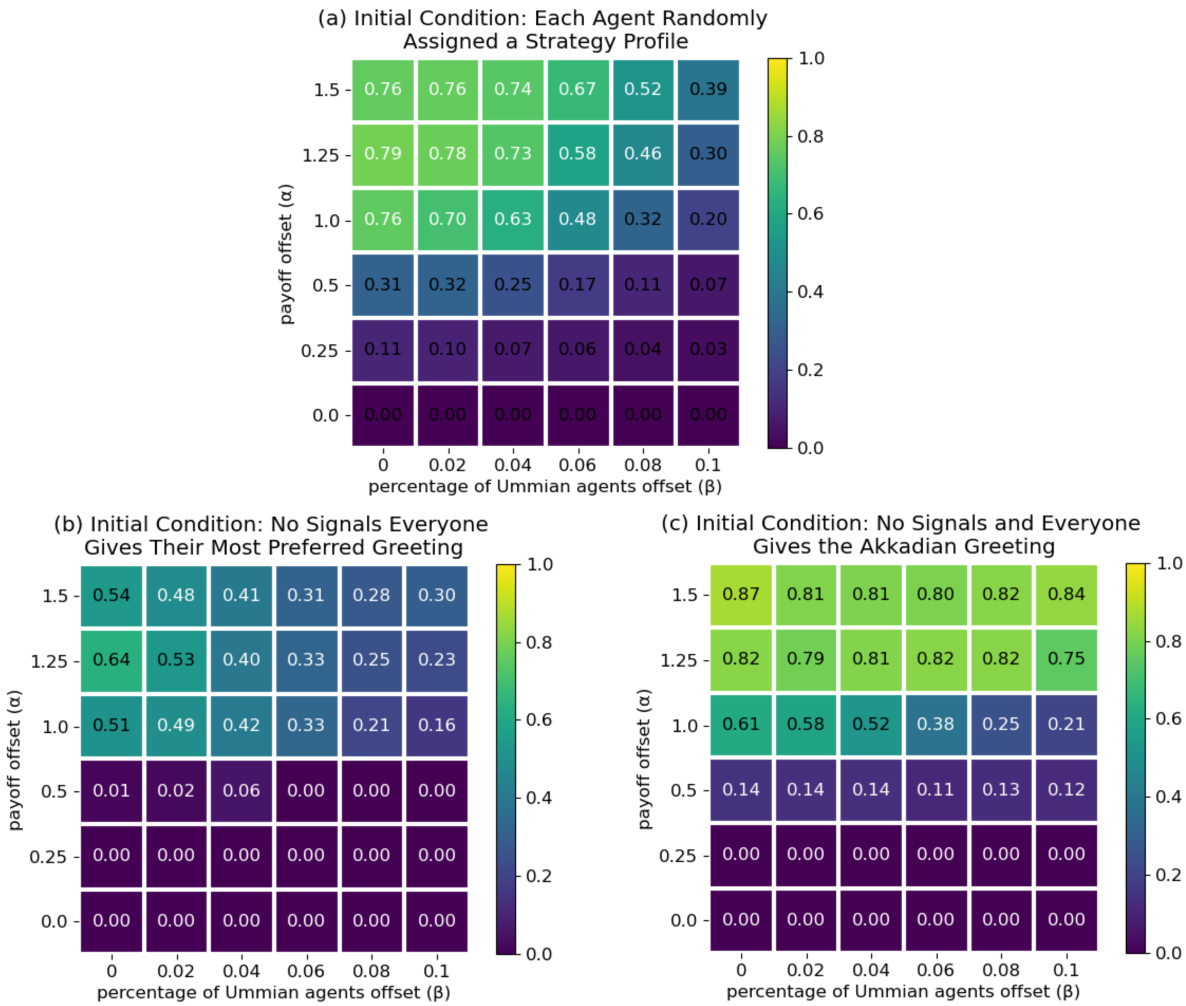


Figure A-17: The proportion of simulations that resulted in the optimal outcome for initial conditions in which (a) agents are randomly assigned a strategy profile, (b) every agent gives their preferred greeting and nobody signals, and (c) every agent gives the Akkadian greeting and nobody signals.

Appendix I Intersectional Preferences with One Dimension of Signals

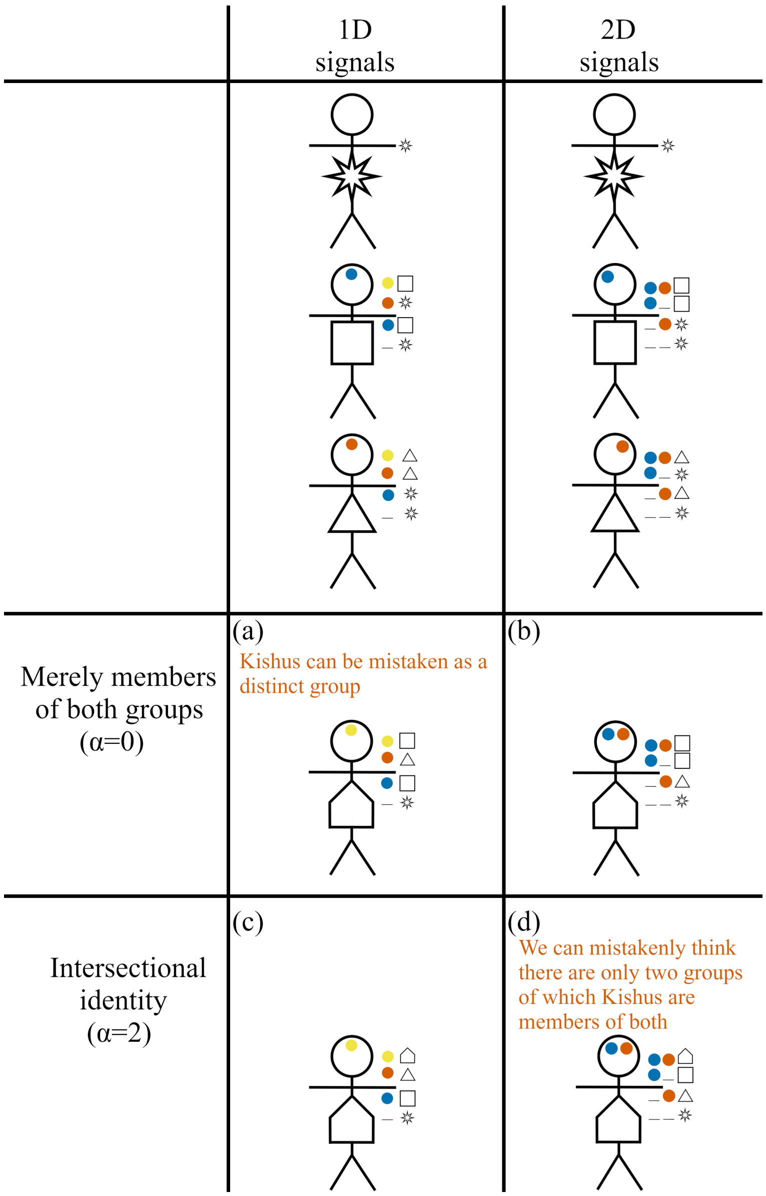
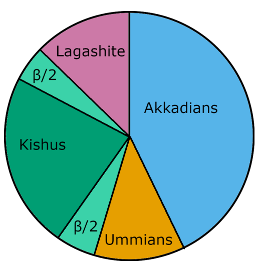


Figure A-18: An illustration of two scenarios in which social signals are misleading if agents actions are not also considered.

This appendix explores parameter space of the intersectional structures, but allows agents three signals in one dimension rather than one signal in each of two dimensions (see Figure A-18). The one dimensional signaling model is capable of, in a sense, producing the same optimal outcomes as the two dimensional version. However, these optimal outcomes are significantly less frequent than in the model with two dimensions of signaling. Where the two dimensional model produced the appropriate signaling system nearly 100% of the time, the one dimensional model (see Figure A-19) achieves the appropriate signaling system much less frequently. The most plausible explanation seems to be that when using multiple dimensions of signals, Lagashites and Ummians do not have to learn anything about the Kishus for the Kishus to successfully coordinate on the optimal action. Kishus can simple use the same signal as either Lagshites and Ummians in corresponding dimensions and to those agents Kishus look like members of their own group. When there is only one dimension of signals, then there is added difficulty for how various types of agents learn optimal behavior because now an optimal outcome requires not only that Lagashites and Ummians learn how to signal their own identities and condition their actions appropriately, but also have to learn what the Kish social signal is and what the optimal action when encountering that signal is. In the one dimensional system, Kishus do not look like Lagashites or Ummians from the perspective of those two groups and it is consequently a more difficult learning problem.

Coordination preferences were equivalent to or close to those given in Section 4.3, i.e.:



Coordination Preferences	Umma greeting	Kish greeting	Lagash greeting	Akkadian greeting
Ummians (□)	1	0	0	0.25 or 0.5
Kishus (⌂)	1	α	1	0.25 or 0.5
Lagashites (△)	0	0	1	0.25 or 0.5
Akkadians (✲)	0	0	0	1

In Section 4.3 Ummians and Lagashites get a payoff of 0.5 for coordinating on the Akkadian greeting and the evolutionary dynamics used $m_0 = 0.01$ and $m_1 = 0.1$. In this appendix we additionally consider, moth those values and a payoff of 0.25 for the Akkadian greeting as well as $m_0 = 0.05$ and $m_1 = 0.2$. The one dimensional signaling system from Figure A-18 is a prerequisite for th optimal outcome to obtain. Figure A-19 shows the prevelance of that signaling system. This contrasts starkly with the two dimensional model in Section 4.3, for which optimal signaling occurs in close to 100% of simulations. Figures A-20 & A-21 further illustrate the extent to which Kishus rarely give the Kish greeting among themselves, despite this being an optimal outcome when $\alpha = 2$.

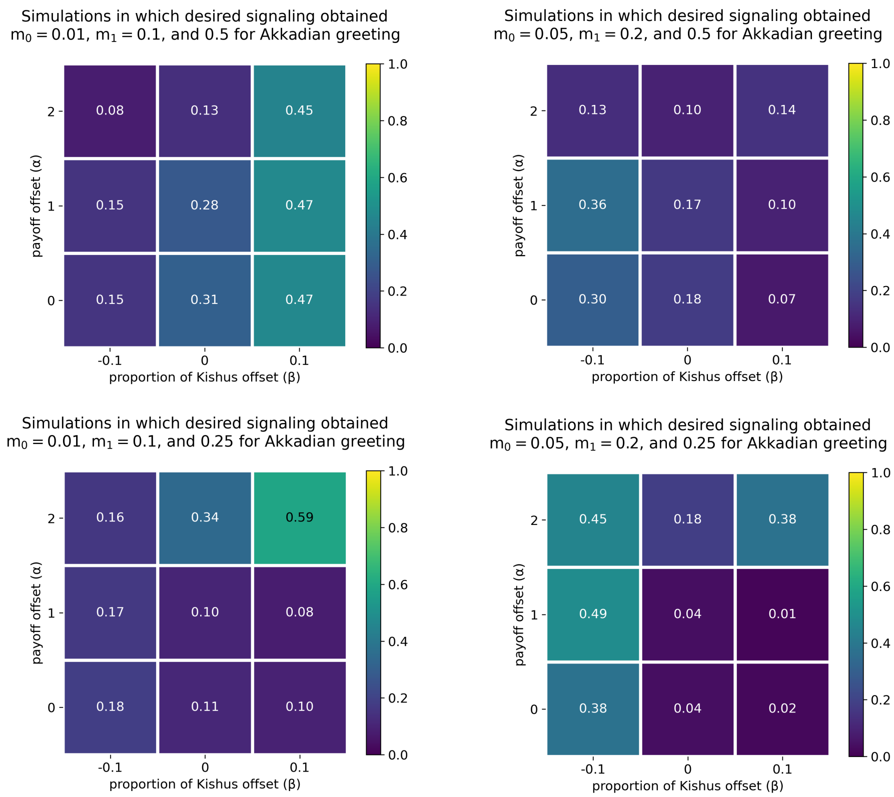


Figure A-19: Prevelance of the optimal one dimensional signaling system. This figure is the one dimensional equivalent of Figure ??

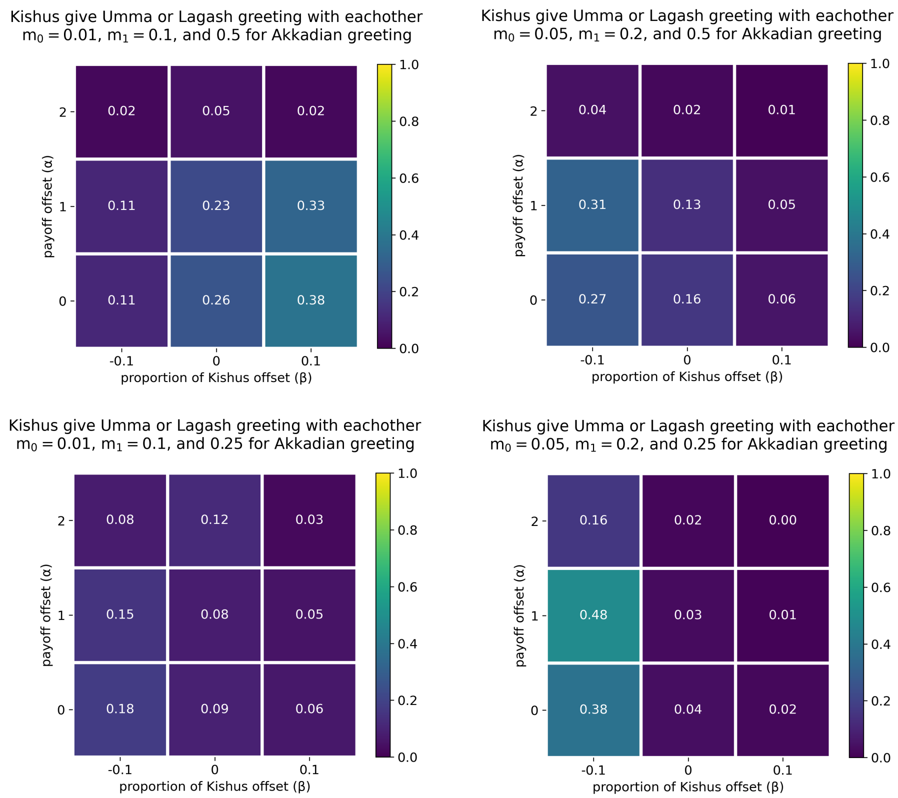


Figure A-20: This figure is the one dimensional equivalent of Figure 11(a), the proportion of outcomes in which Kishus (⌂) give the Kish greeting with eachother in addition to exhibiting the desired signaling behavior.

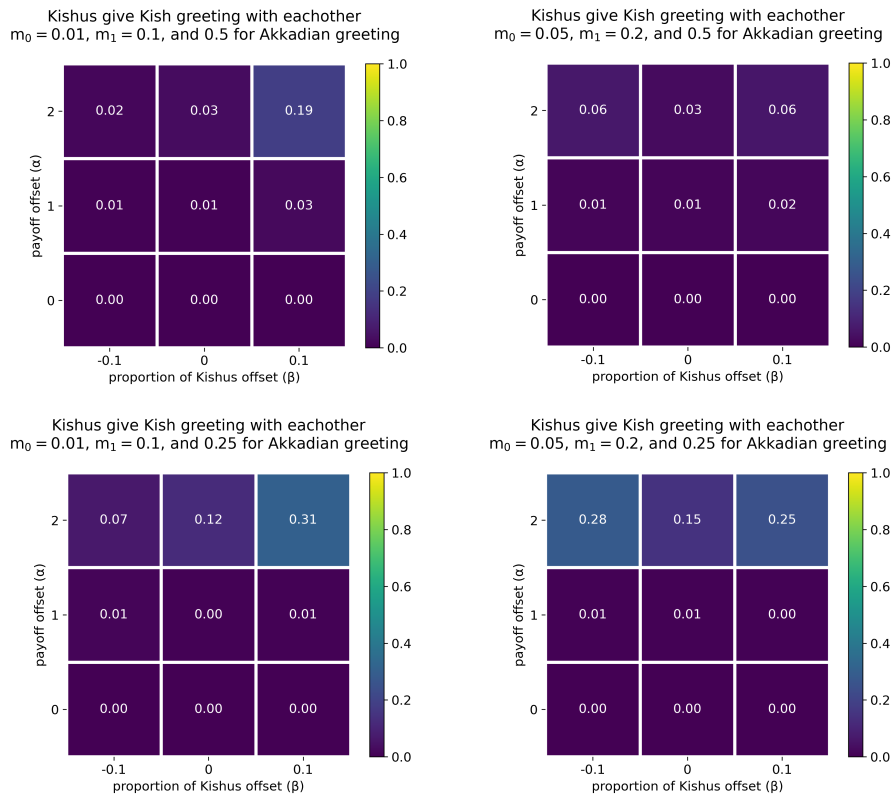


Figure A-21: This figure is the one dimensional equivalent of Figure 11(b), the proportion of outcomes in which Kishus (⌂) give the Umma (□) or Lagash (△) greeting with eachother in addition to exhibiting the desired signaling behavior.

Appendix J Intersectional Preferences with Common Middle Ground

In the main body of the paper, the section on intersectional social identities (Section 4.3) assigned Ummians and Lagashites preferences such that only their own city's greeting and the Akkadian greeting. This was appropriate for illustrating the two very different Kishu identities that can occur for the same signaling system. However, it is a suboptimal choice for thinking about group power dynamics under the same paradigm as prior sections of the paper. In prior sections, a group's power was discussed in terms of group A exerting their power over group B by making group A's preferred action the norm when agents from the two groups interact. If Ummians and Lagashites have zero payoff from the Kish greeting, then it is apriori determined that they will not give the Kish greeting when interacting with Kishus. This appendix shows simulation results from two different amendments to the coordination preferences that give Ummians and Lagashites non-zero payoff for the Kish greeting. It also shows along side these results, comparable simulation results for coordination preferences like the original simulations where they have no payoff for the Kish greeting, though we also vary Ummians and Lagashites' payoff for their most preferred greetings. Note that, unlike Section 4.3, this appendix only considers coordination preferences in which Kishus have a higher payoff for their own greeting than for any others. So, the optimal outcome is always one in which Kishus give the Kish greeting among themselves.

Figures A-26 - A-29 show that, inline with expectations from prior sections of the paper, decreasing the size of the intersectional group makes it less likely that their preference will be the norm when interacting with agents who are members of just one of the two groups that comprise the intersectional group. Increasing the size of the intersectional group (i.e. increasing β) makes it more likely, though still a small likelihood, that the intersectional (Kish) greeting is the norm for interactions between agents in the intersectional group and agents who are members of just one of the two groups. Figures A-23 - A-25 are the measures that were used in Sections 4.3; and are included merely for comparability with that section.

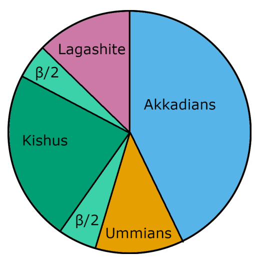


Figure A-22: Proportion of agent types

Left (a) Coordination Preferences	Umma greeting	Kish greeting	Lagash greeting	Akkadian greeting
Ummians (□)	$1+\alpha$	0	0	0.5
Kishus (⌂)	1	$1+\alpha$	1	0.5
Lagashites (△)	0	0	$1+\alpha$	0.5
Akkadians (✲)	0	0	0	1

Center (b) Coordination Preferences	Umma greeting	Kish greeting	Lagash greeting	Akkadian greeting
Ummians (□)	$1+\alpha$	0.5	0	0.5
Kishus (⌂)	1	$1+\alpha$	1	0.5
Lagashites (△)	0	0.5	$1+\alpha$	0.5
Akkadians (✲)	0	0	0	1

Right (c) Coordination Preferences	Umma greeting	Kish greeting	Lagash greeting	Akkadian greeting
Ummians (□)	1+α	0.5	0.5	0.5
Kishus (⌂)	1	1+α	1	0.5
Lagashites (△)	0.5	0.5	1+α	0.5
Akkadians (✵)	0	0	0	1

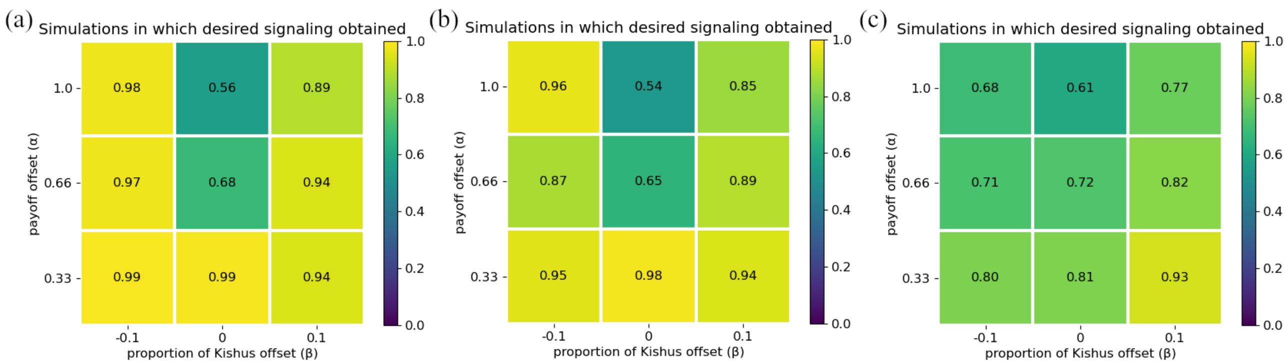


Figure A-23: How often the intersectional signaling system obtains.

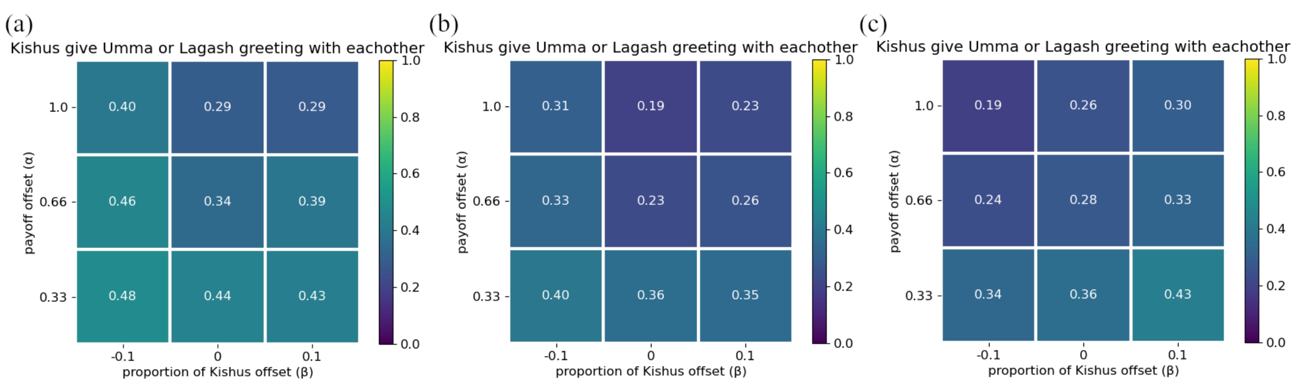


Figure A-24: How often Kishus do not give the Kish greeting among themselves among themselves.

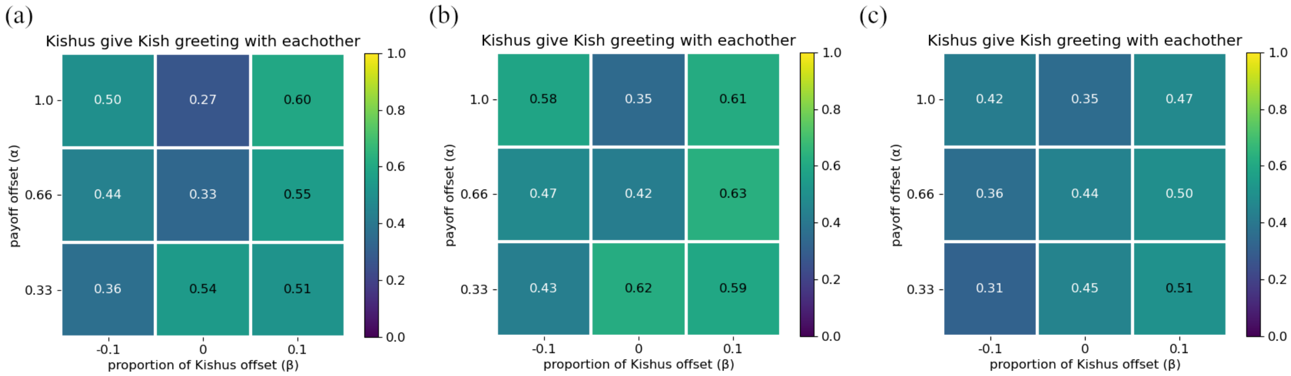


Figure A-25: How often Kishus give the Kish greeting among themselves among themselves.

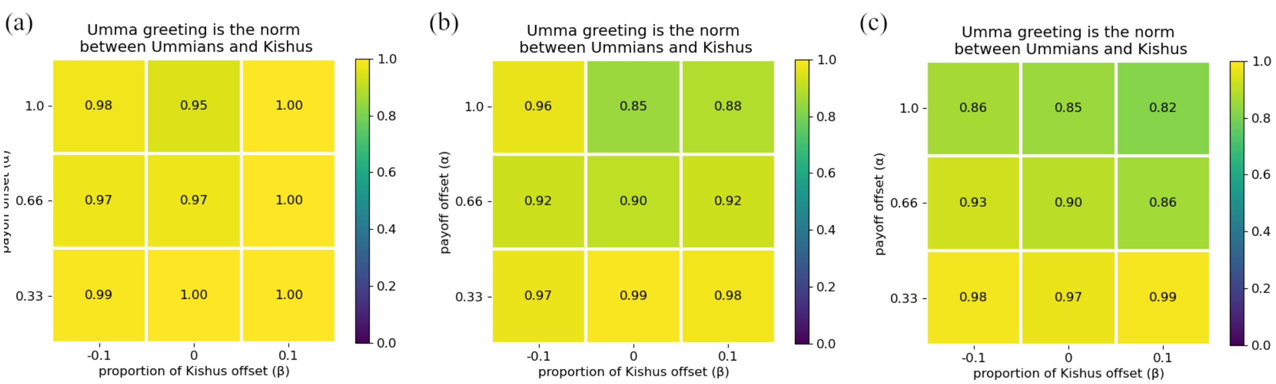


Figure A-26: How often Umma greeting is the norm between Ummians and Kishus.

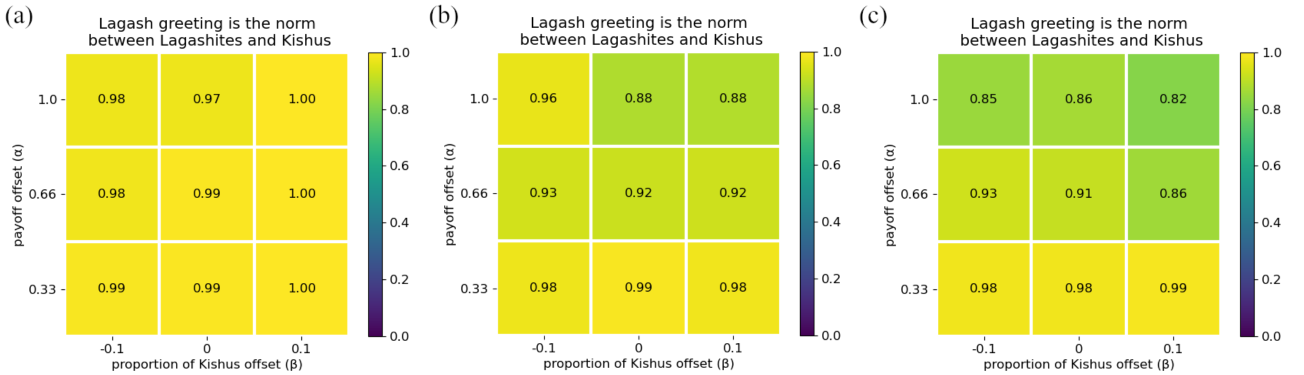


Figure A-27: How often Lagash greeting is the norm between Lagashites and Kishus.

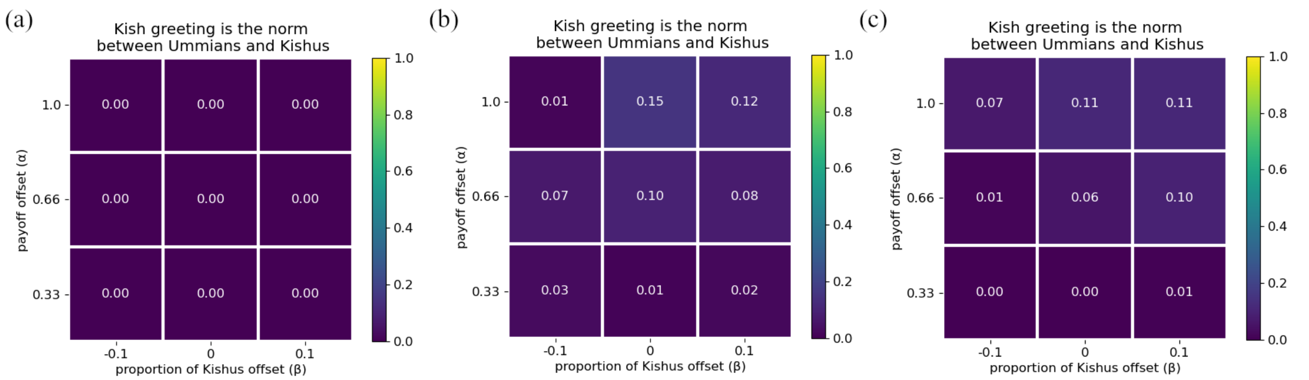


Figure A-28: How often Kish greeting is the norm between Ummians and Kishus.

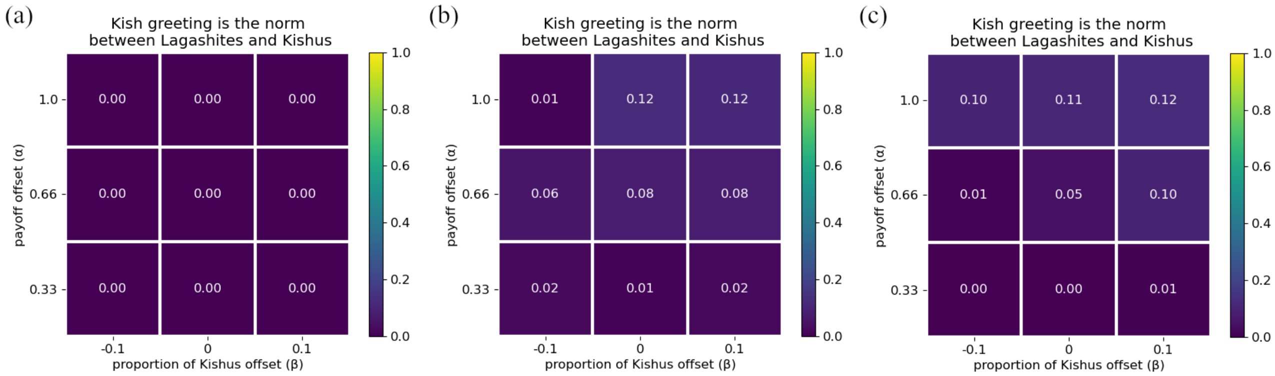


Figure A-29: How often Kish greeting is the norm between Lagashites and Kishus.

Appendix K Adding in Partial Payoffs on Failures to Coordinate

This appendix shows simulation results for the model with one change. The change is that instead of having $p_{xi} = 0$ when strategy profiles x and i result in a failure to coordinate p_{xi} is twenty percent of the payoff that would have been received if the action chosen by profile x had lead to successful coordination. For example, if an agent receives a payoff of 1 for coordinating on a particular action, then if that agent performs the same action but fails to coordinate her payoff will be 0.2. Of course this value of twenty percent was not hard coded and is a parameter that can be varied. But, twenty percent is sufficient for illustrating what happens in the model. If the payoff percentage on failures to coordinate is too small, then simulation results are identical to results from the model in which agents get nothing on failures to coordinate; and, if the payoff percentage on failures is too high, then agents will always perform their most preferred action because the payoff on failures for that action exceeds their payoff for successful coordination on a less preferred action.

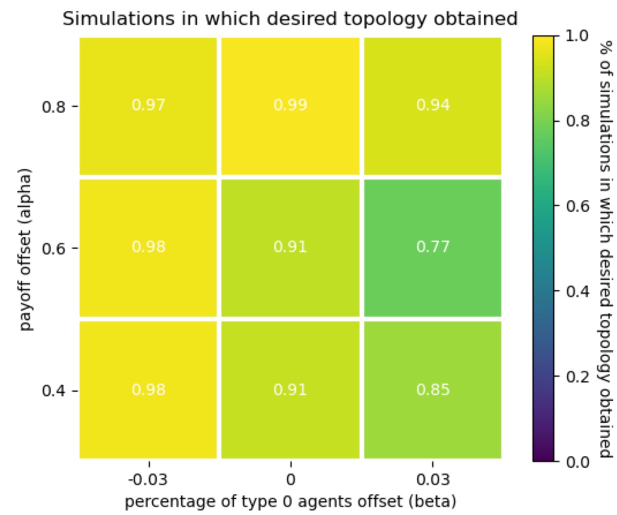


Figure A-30: Proportion of outcomes that are the single embedding structure under parameters identical to those used for Figure 9 (c) but with the modified model where agents get twenty percent on failures.

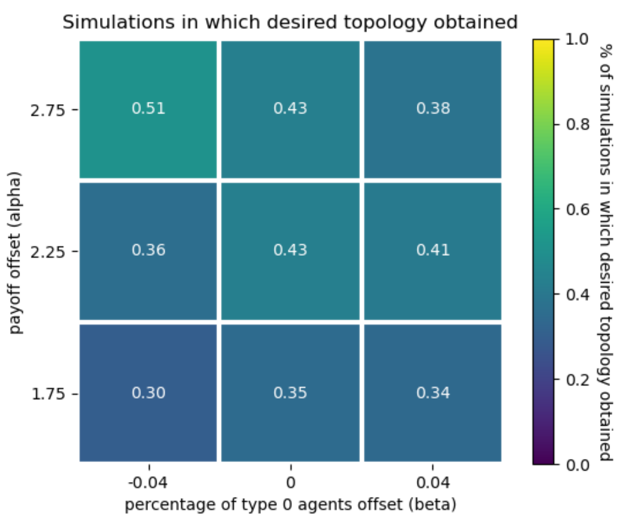


Figure A-31: Proportion of outcomes that are the disjoint double embedding structure under parameters identical to those used for Figure 12c but with the modified model where agents get twenty percent on failures.

Appendix L Analysis for 2x2x2 game

2 types, 2 signals (α, β), and 2 actions (no homophily)

In this simple system, this appendix analyzes when it is the case that an agent should play their less preferred action.

		Type 1	
		Bach	Stravinsky
Type 0	Bach	a, b	c, c
	Stravinsky	c, c	b, a

With $a > b$ and $c < 0$.

- Let $n_i, i \in \{0,1\}$ the proportion of the population that is type i.
- Let $x_{ijk}, i \in \{0,1\}, j \in \{\alpha, \beta\}, k \in$ {Bach, Stravinsky} be the probability that a type i agent plays k given signal j is observed.
- Let $y_{ij}, i \in \{0,1\}, j \in \{\alpha, \beta\}$ be the probability that a type i agent transmits signal j. Note that $y_{i\beta} = 1 - y_{i\alpha}$
- Let A_i be type i's preferred action and B_i be the alternative action.
- Let $p(i|j), i \in \{0,1\}, j \in \{\alpha, \beta\}$ be the probability that signal j was sent by a type i agent given signal j was observed.
- Let $E_{ij}(k), i \in \{0,1\}, j \in \{\alpha, \beta\}, k \in$ {Bach, Stravinsky} be the expected utility for a type i agent playing k after having seen signal j.

Then:

$$p(i|j) = \frac{n_i y_{ij}}{n_0 y_{0j} + n_1 y_{1j}}$$

$$\begin{aligned}
E_{ij}(A_i) &= a[y_{i\alpha}(x_{0\alpha A_i} p(0|j) + x_{1\alpha A_i} p(1|j)) + (1 - y_{i\alpha})(x_{0\beta A_i} p(0|j) + x_{1\beta A_i} p(1|j))] \\
&\quad + c[y_{i\alpha}(x_{0\alpha B_i} p(0|j) + x_{1\alpha B_i} p(1|j)) + (1 - y_{i\alpha})(x_{0\beta B_i} p(0|j) + x_{1\beta B_i} p(1|j))] \\
E_{ij}(B_i) &= b[y_{i\alpha}(x_{0\alpha B_i} p(0|j) + x_{1\alpha B_i} p(1|j)) + (1 - y_{i\alpha})(x_{0\beta B_i} p(0|j) + x_{1\beta B_i} p(1|j))] \\
&\quad + c[y_{i\alpha}(x_{0\alpha A_i} p(0|j) + x_{1\alpha A_i} p(1|j)) + (1 - y_{i\alpha})(x_{0\beta A_i} p(0|j) + x_{1\beta A_i} p(1|j))]
\end{aligned}$$

In the initial state of the reinforcement learning model, $x_{ijk} = y_{ij} = 0.5$ for all i, j, and k. Since $a > b$, it follows that in the initial state of the game agents should always play their preferred action; i.e. $E_{ij}(A_i) > E_{ij}(B_i)$ for all i and j.

Given that all agents have more incentive to play their preferred action than the alternative action irrespective of the signal let's assume that $x_{ijA_i} = 0.5 + \epsilon$ for all i and j where $\epsilon \geq 0$. (Note that $x_{ijB_i} = 1 - x_{ijA_i}$). Furthermore, let's assume that due to random drift $y_{0\alpha} = y_{1\beta} = 0.5 + \delta$ for some $\delta > 0$. Without loss of generality, consider expected utility for type 0 agents. We the get that type 0 agents have higher expected utility for playing their less preferred action when observing β (the more frequent signal among type 1 agents) if $E_{0\beta}(B_0) > E_{0\beta}(A_0)$.

Let:

$$Q = \frac{n_0(.5 - \delta)}{n_0(.5 - \delta) + (1 - n_0)(.5 + \delta)}$$

$$R = \frac{(1 - n_0)(.5 + \delta)}{n_0(.5 - \delta) + (1 - n_0)(.5 + \delta)}$$

If $2Qc - 2Rc - Qb + Rb - Qa + Ra > 0$, then it can be checked that $E_{0\beta}(B_0) > E_{0\beta}(A_0)$ when:

$$\epsilon > \frac{Qa + Ra - Qb - Rb}{4Qc - 4Rc - 2Qb + 2Rb - 2Qa + 2Ra}$$

Here are some plots of:

$$\epsilon = \frac{Qa + Ra - Qb - Rb}{4Qc - 4Rc - 2Qb + 2Rb - 2Qa + 2Ra}$$

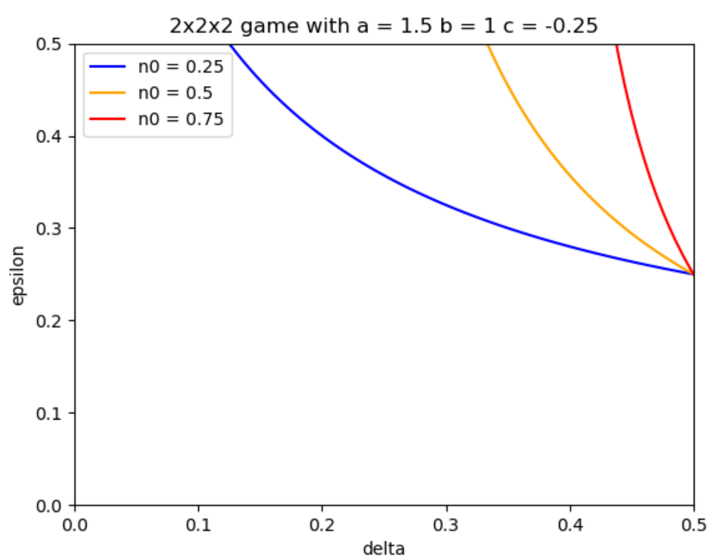


Figure A-32

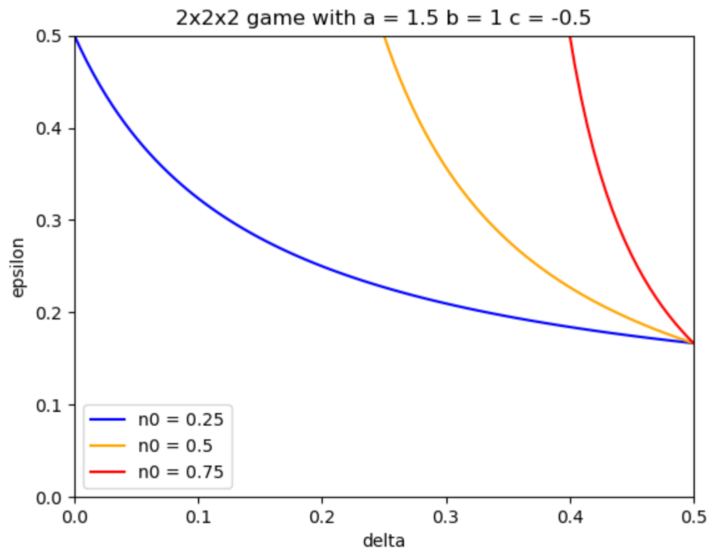


Figure A-33

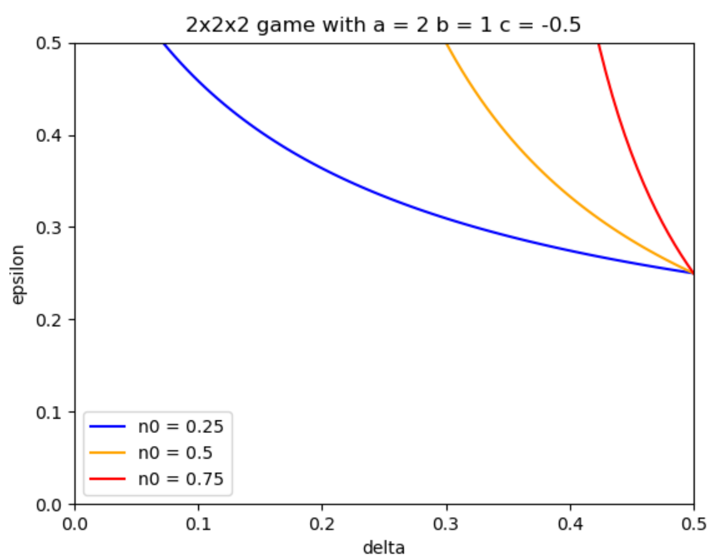


Figure A-34